

User as Engram: Internalizing Per-User Memory as Local Parametric Edits

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Abstract

Personal memory in a language model is two problems, not one: *content*—the specific facts about a user—and *reasoning skill*—the general ability that turns those facts into answers. The brain keeps the two apart—a sparse, local trace (an *engram*) in the hippocampus for each new episode, a slow neocortex for the shared skills that interpret it—so a new fact need not overwrite everything else. The dominant recipe for LLM personalization, the per-user LoRA adapter, does the opposite: it folds content and skill into one global weight delta, and pays for it. Writing a user’s facts as a per-user LoRA contaminates text that has nothing to do with the user; writing the same facts as local *Engram rows* leaves it mathematically untouched—a roughly $34,000\times$ smaller footprint on unrelated text, by design.

We therefore propose **User as Engram**: store a user’s content as surgical edits to the hash-keyed memory table of an Engram model, and carry the reasoning skill in one *shared* adapter that everyone uses—an artificial analogue of the brain’s complementary learning systems. This layered design matches per-user LoRA’s direct recall while delivering $7.4\times$ higher indirect-reasoning accuracy, and never hurts reasoning relative to the untouched base. We can also open the edit up and watch it work on the trained model: writing a fact switches on the memory lookup at exactly its trigger, adds the value the answer needs and nothing else—it leaves every other position unchanged down to the last bit—and stops working if we write it into the wrong layer. Each step is measured, not assumed. Against retrieval, which method wins is decided by how it is deployed, not by how big the benchmark is: a per-user Engram table does not grow with the number of users, so past ~ 100 facts it overtakes a retrieval pipeline running on a $2.5\times$ larger model. The thread tying this together is simple: every method that reaches near-perfect recall pays for it in weaker reasoning, and they differ only in *where* that price is paid—in the shared weights (LoRA), at one address (Engram), or in a search index (RAG). We release four Mini-Engram checkpoints—a nano-scale public replication of Engram—all code and data, and a 50-line multi-tenant server with zero cross-user leakage by design.

Code: <https://github.com/19PINE-AI/user-as-engram>

Website: <https://ring0.me/research/user-as-engram>

1 Introduction

Maya has talked to her AI assistant for the better part of a year. Over those months it has learned that her cardiologist is Dr. Elena Vasquez, that Vasquez practises at Lakeside Cardiology, that Maya is severely allergic to penicillin, and that she went vegetarian in 2024. A useful

assistant must do three things with this *personal memory*: recall a fact on demand (“*who is my cardiologist?*”), reason over it when the question is indirect (“*I’m visiting my daughter in California next month—where should I go for a heart check-up?*”), and never let one user’s facts leak into another’s. Serving millions of such users from a single transformer [Vaswani et al., 2017] backbone is an open architectural problem.

Human memory already solves a version of it. The brain does not record a new fact by nudging every synapse a little; it writes the episode as a sparse, local trace—an *engram*—in the hippocampus, while the slow, distributed neocortex supplies the general skills that interpret it [Semon, 1921, Tonegawa et al., 2015, McClelland et al., 1995]. Keeping the fast, local store apart from the slow, shared one is exactly what lets us learn that Maya is allergic to penicillin without disturbing how we reason about allergies in general.

Personal memory in a language model, we argue, has the same two-part structure—*content* (the user’s specific facts) and *reasoning skill* (turning facts into answers)—and each wants a different kind of memory.

Three families of methods address personal memory, and the dominant parametric one conflates these two parts. In-context learning (ICL, Brown et al. 2020, Min et al. 2022) prefixes the facts; retrieval-augmented generation (RAG, Lewis et al. 2020, Gao et al. 2023) fetches them at query time; per-user adapters bake them into the model’s weights.

This parametric family has converged on one recipe—a per-user LoRA [Hu et al., 2022] on a frozen base’s attention projections, fit on each user’s facts (Section 2)—which we adopt as our reference **per-user-LoRA** baseline. A LoRA is one change layered over the whole model: it keeps a user’s facts and the reasoning skill in the same parameters and, like nudging every synapse, changes the model everywhere, not just where the user’s facts are relevant.

Why a per-user LoRA’s edit is global. This is not a side-effect of how we trained it. It is the price of keeping a fact inside a function the whole model shares. A LoRA has no address: to make one fact more likely, gradient descent has to bend $Q/K/V/O$ in whatever directions fit it most cheaply, and those same directions are also in use on unrelated text—so every forward pass feels the change.

This shows up as two separate costs. The first is extra loss on unrelated text: it is large, and we see it on every base we tried. The second is what the user notices—worse reasoning—and it depends on the base. A heavily instruction-tuned model shrugs the change off; a weaker base LM does not. Earlier work looked only at the second cost on strong bases and concluded the recipe was harmless. The first cost shows the edit was never local (Section 3).

The principle: put each fact where storage belongs. We put the fact where storage belongs—a memory table, not the shared weights. **User as Engram** writes each of Maya’s facts as a surgical edit to the hash-keyed memory table of an Engram-pretrained model [Cheng et al., 2026], whose gated lookups fire only on suffix N-grams via deterministic multiplicative-XOR hashing, while the reasoning skill that interprets the facts lives in one *shared* adapter trained across other users.

Why it works: the table is content-addressable and inert off its key, so writing a fact changes the model’s output only at the positions whose trigger N-gram addresses it—and we can verify the change is *exactly* 0.000 everywhere else, through every subsequent attention and MLP layer, not merely small in the limit (Section 4.3, Figure 5).

Why the table is the right place for personal memory, in three ways the weights cannot offer: writes are isolated, so per-user facts compose and never leak across tenants by construction; the index matches the access pattern, since personal facts are queried by their surface form and a suffix-N-gram hash is a natural key for exactly that; and a row is a fixed ~ 88 KB independent of the model, with a deterministic address that can be swapped in per request, so storage grows with the number of users, not the size of the model.

The same price, paid in different places. The three families are not really competing on quality. Each one reaches near-perfect direct recall, and each one pays for it with weaker reasoning over the stored facts. What differs is *where* the price shows up, and how fast it grows as you add facts and users. A per-user LoRA pays in the shared weights, so the cost is there on every query and adds up as users are stacked together. An Engram row pays at the one place its fact is stored, so the cost grows only with how many facts a single user has. Retrieval pays in the search step, where the cost grows with the size of the pool the encoder must search. Asking *where the price is paid* is what turns “which method is better” into a question about how the system will be deployed—the view the rest of the paper makes concrete.

Contributions.

1. **The cost of storing facts in shared weights—and why earlier work missed it** (Section 3). A per-user LoRA has no address: to fit even one fact, gradient descent must bend the same $Q/K/V/O$ directions that are active on unrelated text, so the edit changes every forward pass—it contaminates globally. An Engram row, by contrast, fires only on its trigger N-gram. On a controlled base this is a $\sim 34,000\times$ difference in extra loss on unrelated text—a property of how a LoRA stores a fact, not of how we tuned it. This extra loss is always there; what a user actually notices—worse reasoning—only shows up on a weak base. A strong, instruction-tuned base answers the question on its own and hides the damage, which is exactly why earlier work judged the recipe harmless. We show you can turn the visible damage on and off just by changing how strong the base is.
2. **User as Engram, with the mechanism opened up** (Section 4): a way to write a user’s facts as small, local edits to an Engram model’s memory table. Unlike a LoRA, whose change is spread invisibly across the whole model, every step of an Engram write can be watched on the trained model: it switches on the memory lookup at exactly its trigger, adds the value the answer needs and nothing else (every other position is unchanged to the last bit), and stops working if written into the wrong layer (Figures 5–6). We release four Mini-Engram checkpoints (178 M–1.22 B; a nano-scale public replication of Engram), all code and data, and a 50-line multi-tenant server with zero cross-user leakage.
3. **A layered design** (Section 6) that pairs per-user Engram content with one shared reasoning adapter—an artificial complementary learning system—and beats every all-in-one baseline on direct recall, indirect reasoning ($7.4\times$ over per-user LoRA), contamination, and per-user reasoning regressions. The lead survives a personal $\rightarrow$ medical schema shift.
4. **A deployment-scale comparison with retrieval** (Sections 6.4–6.5): which method wins is set by facts-per-user and population size, not benchmark size. Because a per-user Engram table does not grow with the population, it overtakes a strong retrieval pipeline—running on a $2.5\times$ larger model—once the knowledge base passes ~ 100 facts.

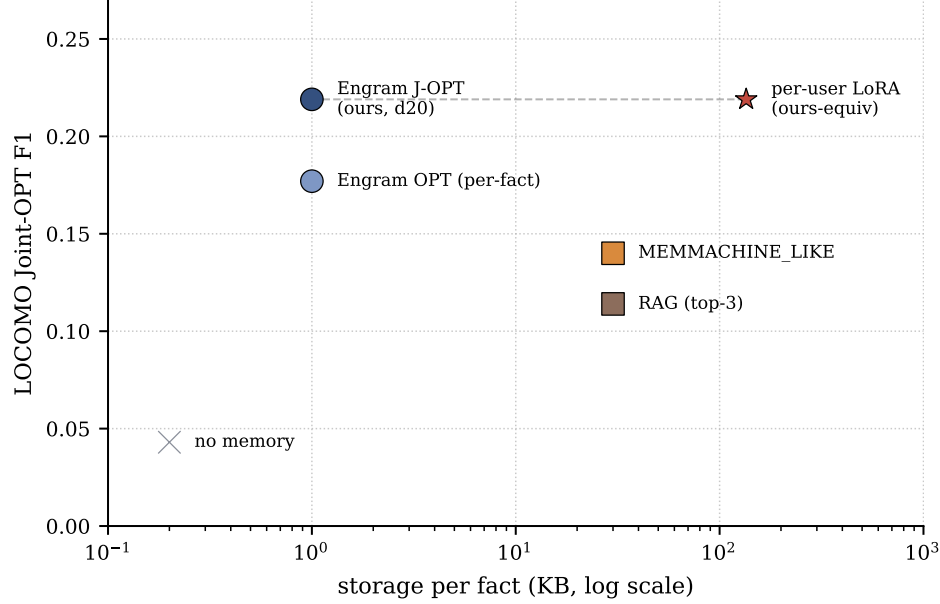


Figure 1. Cost-quality trade-off across personal-memory methods at the same answering LM (Mini-Engram-d20@1536). User-as-Engram Joint OPT matches a per-user LoRA’s LOCOMO F1 at $161\times$ smaller per-user storage (88 KB vs. 14.2 MB at 100 facts). Retrieval baselines sit on the steep low-storage side of the curve but cap out below Engram on quality.

2 Background

The Engram architecture. Cheng et al. [2026], part of the DeepSeek line of sparse models [Dai et al., 2024, DeepSeek-AI, 2024], introduce *conditional memory* as a form of sparsity that complements mixture-of-experts (MoE, Shazeer et al. 2017, Dai et al. 2024). At each token position t , suffix N-grams of canonicalised tokens are hashed via K multiplicative-XOR heads into prime-sized embedding tables $E_{n,k}$. Retrieved rows e_t are projected by learned W_K, W_V and gated by an attention-style scalar $\alpha_t = \sigma(\text{RMS}(h_t)^\top \text{RMS}(W_K e_t) / \sqrt{d})$. The output $\alpha_t W_V e_t$ is added to the residual stream at the selected Engram layer. Crucially, the address $z_{t,n,k}$ is deterministic from the input token IDs—known before the forward pass—so the table can be offloaded to host DRAM without GPU contention.

LoRA-as-memory family. Per-user/per-document LoRA adapters [Su et al., 2025, Tan et al., 2024, Zhuang et al., 2024, Charakorn et al., 2025, Tan et al., 2024b, Bini et al., 2025] train a low-rank delta [Hu et al., 2022, Houlby et al., 2019, Mangrulkar et al., 2022]—one member of a broad parameter-efficient fine-tuning family that also includes prefix-tuning [Li & Liang, 2021] and adaptive-rank and quantized variants [Zhang et al., 2023, Dettmers et al., 2023]—on $Q/K/V$ (and sometimes MLP) matrices on top of a frozen base. It is fit per user via NTP on a (observation, fact, QA) mixture, via knowledge distillation from a teacher (DyPRAG [Tan et al., 2025], DistilledPRAG [Chen et al., 2025]), or via hypernetwork emission (T2L [Charakorn et al., 2025]). All variants *edit the model weights globally*: every forward pass sees the LoRA’s change.

Memory architectures and adjacent work. Engram [Cheng et al., 2026] sits in a long line of trainable key-value memory modules [Geva et al., 2021, Lample et al., 2019, He, 2024, Berges et al., 2025, Huang et al., 2024b, 2025, Yu et al., 2025, Pagnoni et al., 2025, Liu et al., 2025], but

none of them addresses writing a single user’s fact in at inference time. Stacking edits without a combiner echoes LoRA stacking for Stable Diffusion [CivitAI, 2024] (LoRAHub [Huang et al., 2023] instead trains one). A mirror-image line keeps memory *outside* the weights but executable—User as Code [Li, 2026] compiles a user’s history into typed state and rules—which we contrast with our in-weights edits in Section 5.

Memory systems for LLM agents. A large body of work gives an agent long-term memory by keeping facts *outside* the weights and retrieving them at query time: paging systems [Packer et al., 2023]; extraction- and graph-based stores [Chhikara et al., 2025, Xu et al., 2025, Rasmussen et al., 2025, Li et al., 2025, Hu et al., 2026, Wang et al., 2026]; and a line that learns *how* to read and write the store [Yan et al., 2025, Yu et al., 2026, Wang et al., 2025], with surveys cataloguing the space [Zhang et al., 2024, Wu et al., 2025, Du, 2026]. None writes a fact *into* the model. (One concurrent system even shares our name [Patel & Patel, 2025] while doing the opposite—it orchestrates memories in natural language.) We use Mem0- and MemMachine-style retrieval as our strongest baselines (Section 5.4) and ask a different question: *when* does writing a fact into the weights beat keeping it in a retrievable store [Pollertlam & Kornsuwannawit, 2026]?

Personalizing language models. Adapting a model to an individual user is an active area in its own right [Zhang et al., 2024, Liu et al., 2025, Xu et al., 2026], with benchmarks such as LaMP [Salemi et al., 2024] measuring personalized generation. Most methods personalize through the prompt or a retrieval store; the per-user LoRA line above is the parametric alternative, and User as Engram is our attempt to win the locality of a prompt at the zero-context cost of a weight edit.

Retrieval and non-parametric memory. Retrieval-augmented generation [Lewis et al., 2020, Gao et al., 2023] is the dominant non-parametric route, spanning retrieve-then-read pretraining, large-scale retrieval pretraining, and self-reflective retrieval [Guu et al., 2020, Khandelwal et al., 2020, Borgeaud et al., 2022, Izacard et al., 2023, Asai et al., 2024]. Its quality is bounded by the encoder’s ability to surface the right evidence among many candidates—exactly the failure mode we measure as the knowledge base grows (Section 6.5).

Personal memory benchmarks. Long-term personal memory is evaluated by LOCOMO [Maharana et al., 2024], LongMemEval [Wu et al., 2024], BEAM [Tavakoli et al., 2025] (multi-turn evolving beliefs), and PersonaMem-v2 [Jiang et al., 2025]. We use synthetic per-user fact corpora for the within-user fact-density sweeps (tighter control over the parameter we vary), and LOCOMO (Section 5.5) for end-to-end evaluation against retrieval baselines. We chose LOCOMO over LongMemEval and BEAM because its per-question evidence pointers let us train a per-fact Engram row and a retrieval entry against the same gold sentence—an apples-to-apples “same gold, different method” comparison. LongMemEval and BEAM remain open for future work.

Knowledge editing and recitation. Knowledge editing changes a fact the model already holds—locating and rewriting weights [Meng et al., 2022, 2023, Dai et al., 2022, Yao et al., 2023], or routing around them [Mitchell et al., 2022a,b, Wang et al., 2024]—with ripple-effect benchmarks measuring the fallout [Cohen et al., 2023, 2024, Meng et al., 2022b]. We do the

opposite: we *add* facts the model never saw and want everything else left alone. Recitation methods, which surface a stored fact before reasoning over it [Wu et al., 2025, Sun et al., 2023], we reuse in Section 3.

Complementary learning systems and continual learning. The extra loss a per-user LoRA imposes on unrelated text is a modern instance of *catastrophic interference*: training on something new overwrites what was there before [McCloskey & Cohen, 1989], which decades of continual-learning work fight by protecting important weights [Kirkpatrick et al., 2017] or replaying old data, with the problem persisting for large language models [Shi et al., 2024]. The brain avoids it with a different architecture, and the word *engram* points to how: an engram is the physical trace a memory leaves in neural tissue [Semon, 1921], and *engram cells*—sparse neuron populations whose reactivation triggers recall—sit largely in the hippocampus [Tonegawa et al., 2015], while general skills live in the slow, distributed neocortex. Complementary learning systems theory [McClelland et al., 1995, Kumaran et al., 2016] argues this split is what lets a new episode be written without overwriting old skills. User as Engram is the same division made architectural: the per-user Engram row is the fast, sparse, local trace (the hippocampal engram); the shared LoRA and frozen backbone are the slow, distributed skill (the neocortex); the gated trigger-N-gram lookup completes a memory from a partial cue; and our measured locality ($\Delta\text{bpb} + 0.00005$ on unrelated text, Section 3) is the engineering form of the pattern separation that keeps the two from interfering. Where continual learning asks one set of weights to absorb new facts gracefully, we sidestep the interference by writing each fact to its own address and leaving the shared weights untouched.

3 Per-user LoRA contaminates globally

Weights are an excellent place to keep knowledge—pretraining packs millions of facts into them accurately. They are a poor place to keep *one user’s* facts written as a *per-user weight edit*, and not for lack of capacity. A per-user LoRA edits $Q/K/V/O$ in every layer, so its edit is unconditioned (part of every forward pass) and must be fit per user from a handful of examples. Two problems follow, and we measure both: the edit is global, so it contaminates text that has nothing to do with the user; and the facts, once trained in, are *recalled but not readily reasoned over*. The first is an architectural cost paid on every base; the second matters most for personalization, and is why a fact belongs in a store the shared reasoning can read rather than in a per-user copy of the weights.

3.1 Architectural contamination on the same base (Mini-Engram-d20)

We hold the base fixed (Mini-Engram-d20@1536, 1.22 B parameters) and train a per-user LoRA (rank-64 on $Q/K/V$, 1,500 steps per fact, single-token gold) on the synthetic user fact sets used throughout. For each of 20 test users we measure val_bpb on a held-out 262 K-token ClimbMix shard—text unrelated to the user’s facts—before and after the edit. Per-user Engram-row Joint OPT is measured under the identical protocol on the same base.

Figure 2 shows the gap; the mechanism is the point. A LoRA cannot write a fact without bending a function every input shares—to lift the probability of Maya’s cardiologist it moves $Q/K/V/O$ along directions that also fire on unrelated text—so the loss on held-out text rises (more than *tripling* here), and it rises a little further with each fact added, because every new

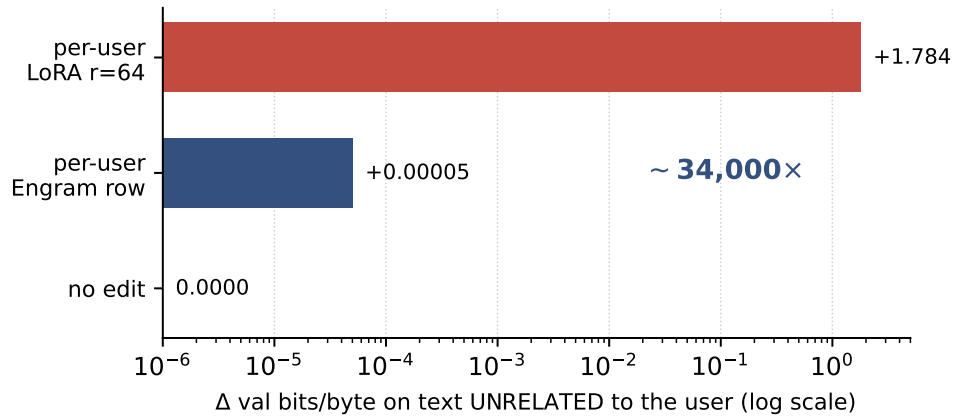


Figure 2. Architectural contamination on held-out text, Mini-Engram-d20 (canonical seed S0, $n=20$ users). A per-user LoRA more than triples val bpb on text *unrelated* to the user’s facts (+1.784; per-user range $[+0.44, +3.74]$; 17/20 users worse), while a per-user Engram row leaves it unchanged to four decimals (+0.00005; 0/20 worse)— $\sim 34,000\times$ less, by design. The 3-seed mean ratio is $\sim 33,000\times$ (Appendix O).

fact bends the function again. An Engram row is written to an address instead: it is read only when its trigger N-gram is hashed, and leaves the loss on everything else unchanged to four decimal places. This is the difference between editing a function and writing to a store, and it is precisely what makes co-locating many users feasible in one and hopeless in the other: contamination from a shared delta accumulates and cannot be quarantined per user, whereas addressed writes have zero cross-talk by construction (Section 7).

3.2 Recall without reasoning

A per-user LoRA learns to *recall* its facts—direct recall is near-perfect—but the harder half of personal memory is *reasoning over* them, and that is where it struggles. The reason is composability. Knowledge acquired in pretraining is laid down in a form the model’s reasoning can chain; a per-user adapter fit to a handful of examples instead memorizes a flat trigger→answer mapping that the shared reasoning does not readily pick up and combine. To answer an indirect question the adapter would have to carry the reasoning skill as well, learned from the same few examples—hard to train, and it does not transfer to facts it never saw. This recall-without-reasoning gap is the recurring finding of the per-user-adapter line [Su et al., 2025, Tan et al., 2024, Zhuang et al., 2024]: the facts go in, but indirect questions over them do not reliably come out.

What a user sees in addition depends on the base. We replicate the per-user LoRA recipe (rank-64 NTP on observation/fact/QA mixtures, 12 epochs $\times$ 200 samples) on four instruction-tuned bases—Qwen2.5-3B and Qwen2.5-7B [Qwen Team, 2025a], Llama-3.1-8B [Grattafiori et al., 2024], and Mistral-7B [Jiang et al., 2023]—and one base LM (Figure 3):

The split is clean (Figure 3). On a base LM the global perturbation also disrupts fragile completion behaviour, so indirect recall drops outright; on an instruction-tuned base the base’s own reasoning carries the indirect question and masks the adapter’s inability to provide it.

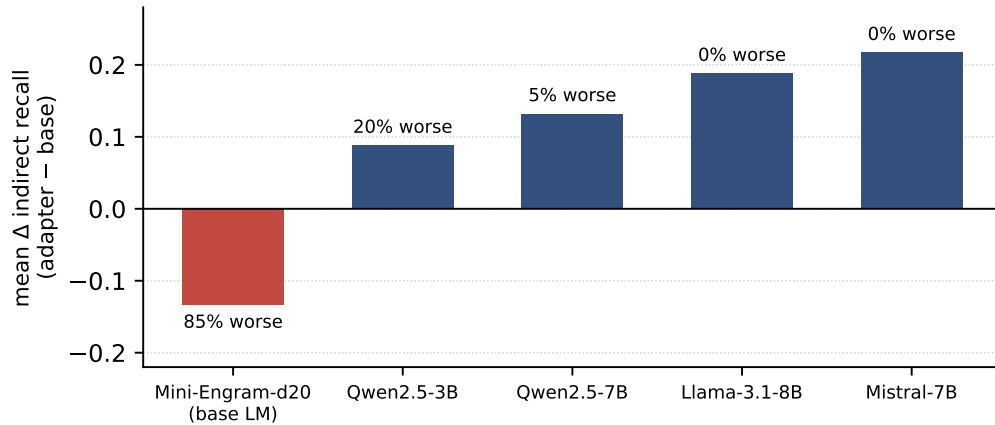


Figure 3. Cross-base LoRA scaling: mean change in indirect recall (adapter – base) with the fraction of users it hurts. On the base LM (Mini-Engram-d20) a per-user LoRA disrupts fragile completion behaviour (85% worse, $\Delta = -0.133$); on every instruction-tuned base the reasoning skill absorbs the perturbation (0–20% worse, Δ from $+0.088$ to $+0.217$).

Why earlier work thought this was harmless. This is worth saying plainly, because it changes how the whole per-user-adapter line should be read. **If you test a per-user LoRA on a strong, instruction-tuned model, you see neither problem: not the extra loss on unrelated text, and not the fact that the adapter never learned to reason over what it stored.** The extra loss is always there, on every base. The thing a user would actually notice—*weaker reasoning*—disappears exactly when the base is strong enough to answer the question by itself. So the recipe looks fine precisely on the models people usually test it on. We can make the damage appear or vanish just by changing how strong the base is (Appendix P.10). The reasoning has to come from somewhere—and Section 6 supplies it from one shared skill, instead of asking every per-user edit to learn it again.

4 Method: User as Engram

We instantiate the “local per-user edit” concretely: surgically write per-user fact rows into the hash-keyed memory table of an Engram-pretrained model. Figure 4 shows where that table sits in the network; the subsections below walk through how a fact is written into it (Appendix Figure 23) and how the writes are served per user (Appendix Figure 24).

4.1 Where the Engram lives

Most of an Engram model is an ordinary transformer [Vaswani et al., 2017]; the only addition is a content-addressable lookup spliced into a few layers (Figure 4; two layers in our Mini-Engrams). At every position, the recent tokens—a suffix N -gram—are hashed by K deterministic multiplicative-XOR heads into addresses in a large embedding table E . The retrieved rows are projected by W_K, W_V , weighted by an attention-style gate $\alpha = \sigma(\cdot)$, and *added to the residual stream*, exactly where attention and the MLP also write. The lookup is thus a read from memory, but keyed by the surface form of the recent tokens rather than by query-key similarity across the sequence.

Two properties make this a genuine store rather than just more parameters, and both are

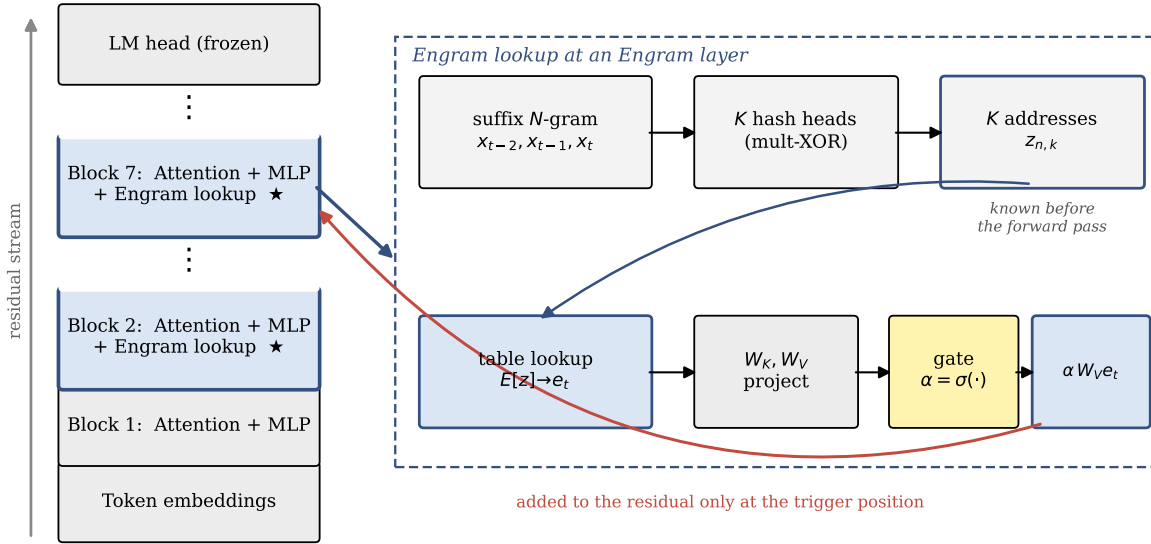


Figure 4. Where the Engram lives in the transformer. The backbone is a standard transformer; at a few designated Engram layers (★) a content-addressable lookup runs alongside attention and the MLP. At a position, the recent tokens’ suffix N -gram is hashed by K heads into K table addresses (deterministic, hence known before the forward pass); the retrieved rows are projected by W_K, W_V , gated by α , and added to the residual stream—but the gate fires only where the trigger N -gram is present. Unlike attention and the MLP, which touch every position and every weight, the Engram read is sparse and addressable; that is what lets a single written row change behaviour locally.

what we exploit. First, the address is a deterministic function of the input token IDs—known *before* the forward pass—so we know exactly which rows a given query will read, and the table can be offloaded to host DRAM. Second, the gate makes the read sparse and conditional: most positions retrieve essentially nothing, and a stored row affects the output only at positions whose N -gram addresses it. This is the addressable, sparse store that per-user facts want—the opposite of the global weight edit of Section 3. Attention and the MLP are dense and global—every position, every weight—whereas the Engram read touches a handful of rows at a handful of positions, so editing one row changes the model only where that row is addressed. The rest of this section turns that into a per-user memory.

4.2 Surgical row insertion

To write a fact, we put its answer where its trigger is read. The trigger’s suffix N -gram hashes to a fixed, sparse set of table rows R_f —about 16 rows for our $N=3, K=8$ configuration—deterministically, touching no other row. What remains is *what* to write into those rows, for which we use three strategies of increasing cost and fidelity (per-strategy recall against a RANDOM control in Appendix D):

- **UNEMBED_P** (closed form): solve in one step for the row value whose projection steers the next-token distribution toward the answer—a single matrix–vector product, <1 ms, no

training.

- **OPT**: refine that value with a few gradient steps per fact (~ 1 s), trading time for accuracy.
- **Joint OPT** (our default beyond ~ 30 facts/user): optimise *all* of a user’s rows together rather than one fact at a time.

Joint OPT is the default because of the one place this design has to earn its keep. When many of a user’s facts are live at once, their rows are read in the same forward passes and interfere; optimising them jointly lets the rows settle into values that coexist, which is what holds recall up as facts-per-user grows (Section 5). Crucially, every write still lands only in that user’s R_f , so however many facts or domains are stacked, the store stays additive and private—the locality is the architecture’s, not a constraint we have to impose.

Read together, the two figures give the method in one sentence: **remembering a fact is writing a few rows**. A fact decomposes into an address set R_f (*where*, fixed by the trigger’s hash) and a value e^* (*what*, set by one of the strategies above), and the write touches nothing else. The locality we rely on is therefore *inherited from the architecture*—the read was already sparse and addressed (Section 4.1)—rather than enforced by a separate mechanism. That is why per-user and per-domain writes compose without a combiner and never leak across users (Section 5), and Section 4.3 verifies that “touches nothing else” is literally exact.

4.3 The local edit is a glass box

A LoRA’s change is spread invisibly across the whole model, so you cannot point to what it did. An Engram write is the opposite: you can watch every step of it on the trained model. We follow one inserted fact through the write and check each step on the trained Mini-Engrams (full detail, with the depth and multi-hop probes, in Appendix B). Three things are worth seeing, and Figures 5–6 show each.

(1) The write switches on its own lookup, and adds just the value it should. A row reaches the output only through a gated value, $\alpha_t W_{ve}$: a switch α_t that decides whether the lookup is read, times the value W_{ve} the row carries. Writing a fact does both at once. The switch at the trigger turns from nearly off to nearly on, while it stays off everywhere else (Figure 6a)—the row we write is also what turns its own lookup on. And the change the row makes to the output points almost exactly along the value it carries (Figure 6b): the switch and the small convolution around it scale the value but do not turn it into something else. The same alignment was reported for the original Engram code on a fresh, untrained model; we confirm it on the trained model, for the strategy we actually deploy.

(2) Nothing else moves—exactly nothing. We write only the trigger’s rows, and the trigger’s last few tokens are unique in the sentence, so every other position reads the rows it always read and its output is unchanged to the last bit. This holds for every one of the 16 test facts and both writing strategies: the largest change at any non-trigger position, at any layer, is 0 (Figure 5a). A LoRA that learns the same fact does the opposite—it moves every position in every layer, and shifts unrelated text as well (Figure 5b). This is the line between editing a function and writing to a store. Note where the locality comes from: not from the switch being off elsewhere, but from the fact that we only ever changed the rows at one address.

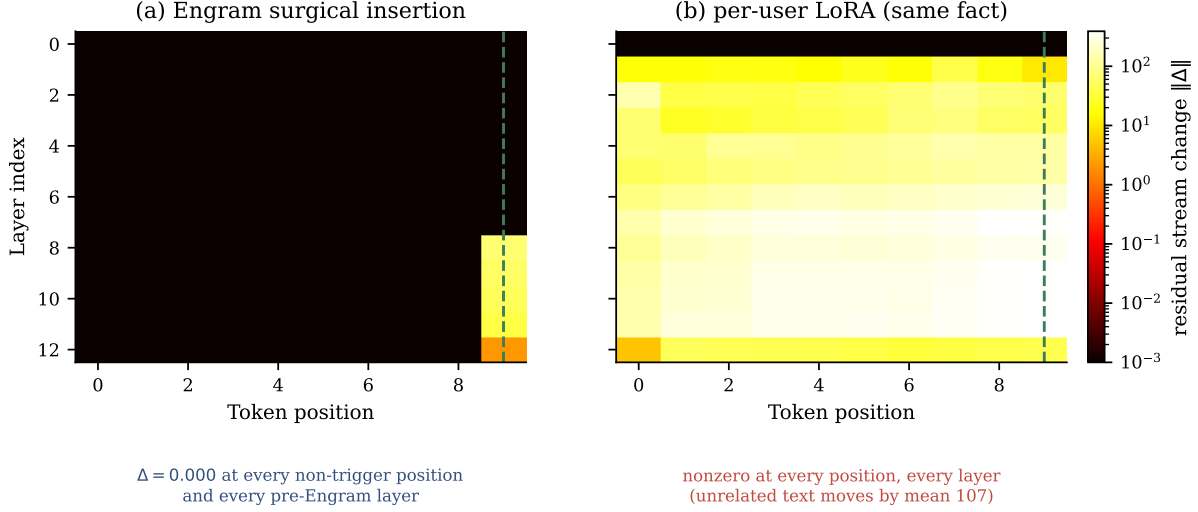


Figure 5. Addressed write vs. global function-bend. Per-layer, per-position residual-stream change $\|x_{\text{after}}^{(\ell)} - x_{\text{before}}^{(\ell)}\|$ on the *same* trigger sentence and base (Mini-Engram-d12@1280, log colour scale). **(a)** An Engram row insertion is *exactly* 0.000 at every position before the Engram layer (causality) and at every non-trigger position after it—only the trigger column moves. **(b)** A per-user LoRA fit to the *same single fact* moves the residual at every position and every layer, and perturbs unrelated text (“The capital of France is”) by mean 107. Storing a fact in a shared function changes the function everywhere; storing it at an address does not.

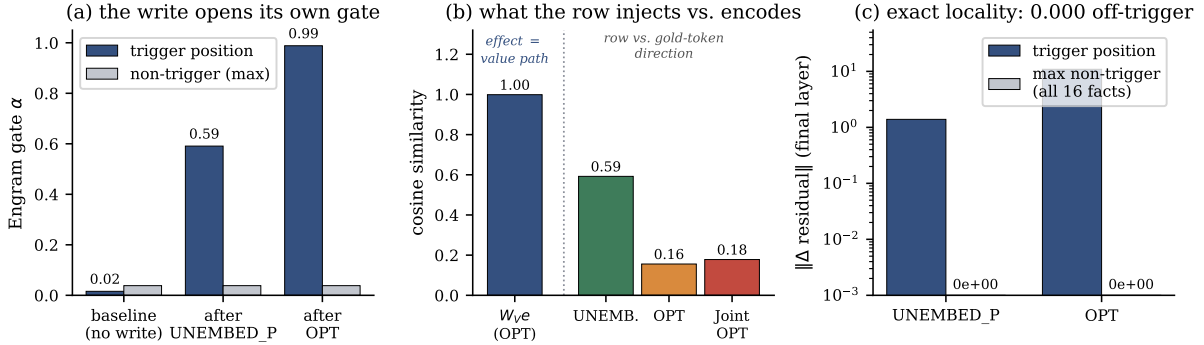


Figure 6. The mechanism on the trained model (Mini-Engram-d20, 16 facts). **(a)** The write opens its own gate: the trigger-position gate α rises from 0.02 to 0.99 (OPT) while non-trigger positions stay at 0.04. **(b)** The deployed row’s residual change is cosine 0.999 to its value-path projection W_{Ve} (left bar); what the row *encodes* relative to the gold token’s unembedding is exact for the closed-form solution (0.59) and drifts as gradient OPT/Joint-OPT trade direct gold-alignment for higher recall (0.16–0.18). **(c)** Exact locality: the maximum non-trigger residual change over all 16 facts is 0.000 for both strategies, versus an order-1 change at the trigger.

(3) It only works near the end of the network. The Engram lookup is read at a late layer, after the model has mostly made up its mind about the next token. That is why a row can change the answer cleanly. To check this is the real cause and not a coincidence, we write the same facts into the model’s *early* lookup instead: at the same effort, recall drops from perfect to about a quarter, because an edit made early gets reworked by all the layers above it. Where the lookup sits is not a free choice—it has to be late enough that the value it adds is close to the final answer.

What this kind of write *cannot* do is join two facts together. The lookup only matches the words it was given, so it answers “who is my doctor” but not “where does my doctor work” unless that second step was stored too (Section 5.6). That is the reason we keep the facts and the reasoning apart (Section 6): the store can hold a fact but cannot chain it, so the chaining has to come from somewhere else. The split between facts and reasoning is forced by how the mechanism works, not a convenience we chose.

4.4 Per-user override tables and additive composition

The production design uses per-user override tables. Each user has a small dictionary $\{r_i \mapsto v_i\}$ of fact rows. At request time, the server saves the originals at the affected addresses, writes the user’s overrides, runs the forward, and restores. Cross-user leakage is *zero by design*—user A’s overrides are not in the table when user B queries.

Override maps with disjoint addresses commute, so corporate facts + user facts (or any number of domain-specific Engrams) **stack additively at inference** without retraining a combiner. This mirrors how Stable Diffusion LoRAs additively stack, but at the row level. Section 5 confirms that additive composition is lossless when the domains have disjoint trigger templates.

5 Experiments

This section answers four questions about User-as-Engram, in order. **(a) How accurately do facts surface?** (Section 5.2) for one fact and for many at once. **(b) What does it cost?** (Section 5.3) storage and composition vs. a per-user LoRA. **(c) Does writing the fact beat retrieving it?** (Sections 5.4–5.5) against memory-system and RAG baselines on the same answering LM. **(d) Where does it break?** (Sections 5.6–5.7) multi-hop chaining and paraphrases. The capacity, token-budget, and dense-size scaling ablations that justify our recipe are in Appendix P. We use three fact corpora throughout: the **base** 200-fact benchmark (100 USER + 100 ORG facts), the **XL** corpus of 1,000 USER + 1,000 ORG templated facts (Section P.4), and the **XXL** corpus of 3,132 distinct trigger templates (high-density and distinct-template stress tests).

5.1 Setup: the Mini-Engram models we insert into

Before measuring per-user insertion we need an Engram backbone to insert into. Cheng et al. [2026] released only architectural reference code; no Engram-trained weights are public. We graft Engram into Karpathy’s nanochat [Karpathy, 2026]—a GPT-2-style backbone [Radford et al., 2019] optimised with Muon+AdamW [Jordan et al., 2024]—and train four Mini-Engram models on a single Blackwell GPU at the ablation-optimal recipe (Section P.3): the same large Engram table ($50\text{ K} \times 256$, 51.2 M params), trained at Karpathy’s 12 tokens-per-param (t/p) budget on the scaling-params for every dense size.

Model-name notation. Throughout the paper we use $d\{\text{depth}\}@\{\text{width}\}$ for the ablation-optimal Mini-Engrams ($d12@768$, $d12@1280$, $d20@1536$) and $v1/v2$ to denote earlier and final training runs at the same shape. Where the width is omitted (e.g. $d12$ without a suffix), we mean the $v2$ (ablation-optimal at width 768, 339 M total) checkpoint. The four headline checkpoints, their param counts and token budgets are in Table 1; all four share the same 51.2 M Engram table.

Table 1. Mini-Engram pretraining at the ablation-optimal recipe. All use the large Engram ($50\text{ K} \times 256 = 51.2\text{ M}$ Engram params). Trained on a single NVIDIA RTX PRO 6000 Blackwell (102 GB) in bf16.

	d8 v2	d12 v2	d12@1280 opt.	d20@1536 opt.
depth $\times$ width	8×512	12×768	12×1280	20×1536
Engram layers	2, 5	2, 7	2, 7	2, 11
total params	178 M	339 M	625 M	1.22 B
scaling-params	42 M	110 M	278 M	617 M
ClimbMix tokens (12 t/p)	0.50 B	1.32 B	3.34 B	7.40 B
final val_bpb	0.912	0.827	0.770	0.730
wall-clock (single GPU)	~ 50 min	~ 5 h	~ 10.5 h	~ 70 h (shared GPU)

We reproduce Engram’s two signature findings. Suppressing the Engram lookup hurts factual recall far more than reading comprehension—the factual-vs-reading sensitivity asymmetry of Cheng et al. [2026] (their §6.3), at the $\sim 10\times$ smaller magnitude expected for a model two orders smaller on a corpus three orders smaller. The Engram layers also act as effective extra depth: a LogitLens probe shows our small Engram model resolving its prediction earlier than its dense-only twin, matching their §6.1 (Appendix Figure 14).

5.2 Per-fact recall and within-user density

A single inserted fact surfaces within ~ 8 points of the in-context ceiling, at three orders of magnitude lower per-user cost than training a full LoRA (Appendix D). The harder question is what happens when many facts are live at once: the rows in a single user’s override table interfere during the forward pass. We sweep that density curve on Mini-Engram-d12 (Appendix Figure 26).

Joint OPT is the recommended default for ≥ 30 facts/user: it tracks multi-fact LoRA rank-64 to within a few top-5 points while keeping top-5 recall above 90% out to ~ 300 facts/user, all at up to $\sim 161\times$ less per-user storage (Appendix Figure 26; per- n recall and storage in Appendix P).

What the density ceiling is—and what it is not. Three ablations triangulate the ceiling to a single cause, ruling out the obvious suspects. (i) It is *not* backbone capacity: holding Engram capacity fixed and scaling the dense backbone does not lift recall at 1 000 facts—it slightly degrades it (top-1 $0.35 \rightarrow 0.28$ from d12@768 to d20@1536, Section 8.3). (ii) It is *not* the table or the hash: per-fact *independent* OPT, where each fact gets private rows, shows *no* ceiling to 1 000 facts (≥ 0.98 top-1, Appendix P.4). (iii) It is *not* pretraining: a multi-fact-in-the-loss finetune only accelerates convergence—given enough OPT steps the baseline reaches the same asymptote (Section P.2). What remains is the one thing all three share: **gradient interference among the co-active rows in a single user’s table, at a fixed inference-time optimisation budget**. This is why the path to breaking it is a recipe change (multi-fact-in-the-loss, scoped tables), not further dense scaling, and why it scales with facts-per-user rather than with the model.

5.3 Cost, storage, and composition

Storage. A user’s whole table is a fixed ~ 88 KB, while a per-user LoRA is 14.2 MB— $\sim 161\times$ larger at 100 facts/user, and $\sim 1700\times$ at 10 facts/user. At a million users that is 100 GB versus 14.2 TB: the difference between one server and a distributed store (Appendix Figure 36). Figure 1 places all the methods on one cost–quality plot: a single fact matches the in-context ceiling at a fraction of LoRA’s cost, and at high density Joint OPT keeps pace with even rank-64 LoRA at tens of times less storage, with the gap turning in Engram’s favour by 1 000 facts.

Composition. Two users’ tables, or a user’s facts and a company’s, occupy different addresses, so they simply add up—no combiner to train. The one exception is when two sets of facts share the same trigger words: their addresses collide and the last write wins. The rule, then, is to add tables when their triggers differ and to keep separate per-user tables when they do not (Appendix F).

5.4 Comparison against memory systems

We compare User-as-Engram against state-of-the-art memory systems that target the same use case (per-user / personal memory). All systems use the *same* Mini-Engram-d12 as the final-answer LM; the only difference is how facts are stored and retrieved. Our MEM0 and MEMMACHINE baselines follow the retrieval recipes of Mem0 [Chhikara et al., 2025] and MemMachine [Wang et al., 2026]. The sentence encoder used by the RAG / MEM0 / MEMMACHINE baselines is all-MiniLM-L6-v2 [Wang et al., 2020, Reimers & Gurevych, 2019] (80 M parameters; comparable in size to a production retriever).

This subsection probes *direct* fact recall under trigger-exact and paraphrased queries; the separate *indirect-reasoning* comparison (does retrieval recover the gold fact across hops, and what happens as the KB grows?) is deferred to Sections 6.4–6.5.

Setup. 100 USER facts (XXL corpus), asked two ways: with the fact’s exact stored prompt (easy for retrieval—the words match), and with a paraphrase (“I love the spice” for “My favorite spice is”; 16 facts $\times$ 4 rewordings).

On exact-trigger queries the words line up with what was stored, so nearest-neighbour retrieval is near-perfect and beats Engram’s 68% top-1—but it pays 16–63 context tokens for the answer, where Engram pays none (Appendix Figure 27).

The story flips on paraphrased queries. When the query is reworded, the sentence encoder confuses surface forms and retrieval slips, while a single-trigger Engram misses the reworded N-gram entirely. The fix is to write a row at every anticipated paraphrase: multi-trigger insertion then wins on paraphrase by a clear margin at zero context cost (Appendix Figure 27), because the contract is explicit—the trigger N-gram—rather than left to an encoder’s notion of similarity. This verdict (*retrieval wins on the verbatim prefix; multi-trigger Engram wins under paraphrase*) is invariant to dense scale: the Engram numbers barely move across our three sizes, and only the retrieval baselines drift as their answering-LM improves (Appendix P).

Our retrieval baselines are the retrieval step itself (top-3 sentence-encoder lookup over atomic facts), not the full Mem0 or MemMachine systems with their extraction and consolidation machinery; we skip those because our facts are already structured [Maharana et al., 2024, Wu et al., 2024].

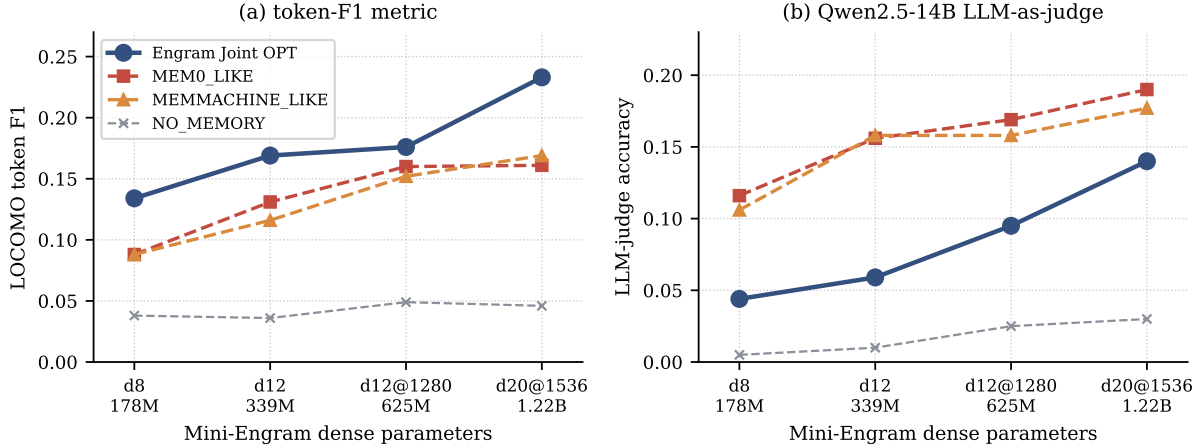


Figure 7. Locomo single-hop, full 10 conversations, scaling with Mini-Engram dense size. **(a) Token-F1:** User-as-Engram Joint OPT (blue) beats every retrieval baseline at every scale. **(b) LLM-judge accuracy** (Qwen2.5-14B-Instruct): the ranking flips—retrieval baselines (MEMO_LIKE, MEMMACHINE) beat Joint OPT by 0.05–0.10 because token-F1 over-credits Engram for the correct first answer token despite a noisy continuation.

5.5 Locomo single-hop fact recall

To evaluate the same memory systems on a published long-term conversational benchmark, we run all eight systems from Section 5.4 on Locomo [Maharana et al., 2024]. We use *all 10 Locomo conversations* and take the first 80 single-hop (non-adversarial) questions per conversation, for a total of ~ 800 question-answer pairs grounded in dialog per model.

Setup. Retrieval baselines store the gold evidence turns; User-as-Engram stores the gold (question, answer) pair. Every system uses the same Mini-Engram-d12 to write the final answer, so the only difference is how the answer is found. We score token-F1 against the gold answer (details in Appendix A).

User-as-Engram Joint OPT clears every retrieval baseline on single-hop token-F1 (Figure 7). Absolute scores are low—Mini-Engram-d12 is a small base LM with no instruction tuning and token-F1 punishes verbosity—so the relative ordering is what matters. This is a best-case-for-each benchmark: retrieval stores the gold evidence sentence, Engram stores the gold (q, a) pair, and the only question is which extracts the answer more reliably under a fixed LM. Engram wins because the gate fires on the question’s trigger N-gram and biases the next token toward the gold answer, whereas retrieval can miss the relevant turn and, even when it hits, the in-context evidence still competes with the base LM’s priors.

Scaling at the optimal config. We repeated the same Locomo setup with all four trained Mini-Engrams at our ablation-optimal recipe (Section P.3); the Engram lead widens monotonically with dense size (Figure 7).

But the win depends on the kind of question. Across Locomo’s four answerable categories, writing the fact wins on single-hop, multi-hop, and reasoning questions—where the stored question $\rightarrow$ answer pair is exactly what is needed—but loses on open-domain questions, where the answer is a long verbatim span of a sentence the model never saw and a one-token nudge cannot rebuild it (Appendix A, Figure 11). So User-as-Engram is a specialised tool, not

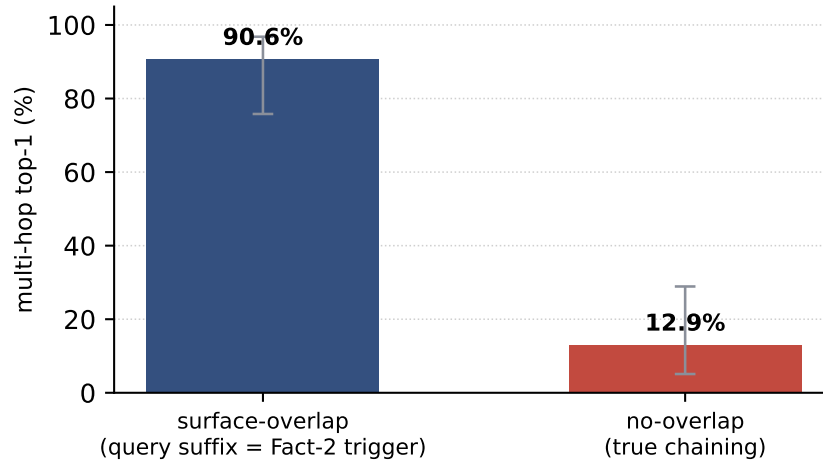


Figure 8. Multi-hop reasoning over Engram-inserted facts ($n=63$ chained pairs on Mini-Engram-d20, two OPT-15 insertions per item). Per-fact direct recall is 99.2% (both facts are insertable), so the gap is chaining, not insertion. When the query’s suffix N-gram coincides with Fact-2’s trigger, the gate fires and recall is 90.6%; when true chaining is required, it collapses to 12.9% (Wilson 95% CIs shown). The gate is a surface-N-gram hash—it matches, it does not compose.

a drop-in replacement for retrieval: a real system would route by question type, or use Engram for the facts it was taught and retrieval for free-form answers from the session.

One caveat about the score, and its fix. Token-F1 gives credit for partial word overlap, and our basic insertion only trains the *first* answer token, so it can score well by getting that first word right over an otherwise noisy answer. When we re-score with an LLM judge (Qwen2.5-14B, Appendix A), first-token Engram indeed slips behind retrieval on single-hop—but training the *whole* answer with multi-token Joint OPT fixes it and again overtakes the best retrieval baseline on the larger models. The paraphrase and storage wins above score only the first token, so they are unaffected. (The judge is stricter than LOCOMO’s official one, so these numbers are not directly comparable to its leaderboard.)

5.6 Multi-hop reasoning over inserted facts

We probe chaining directly: store two facts (“my doctor is Patel”, “Patel works at Globex”) and ask a question that needs both (“where does my doctor work?”). On 63 such pairs, half worded so the question ends in the same words as the second stored fact and half not, we can separate a true chain from a lucky word-match.

The result is almost all-or-nothing (Figure 8): when the question happens to end in the second fact’s words, recall is high; when a real chain is needed, it falls to near chance—and the few hits are word-match coincidences. Since each fact alone is recalled essentially perfectly, the gap is chaining, not storage. This is the limit the store shares with a per-user LoRA: it matches words, it does not reason across them. The recite-then-reason recipe (Section 8) is the natural fix.

5.7 Paraphrase generalisation

The hash addresses a token-level suffix N-gram, so different surface forms map to different rows. The full comparison against retrieval baselines under paraphrase queries is in Appendix Figure 27: multi-trigger insertion reaches 96.9% top-1 vs. MEMMACHINE’s 75.0%, and the lead holds at every size (Section 5.4, “Scale invariance” paragraph). Per-paraphrase rank data are in Appendix L.

6 Layered architecture: shared reasoning skill + per-user content

So far: per-user LoRA contaminates (Section 3) and per-user Engram preserves direct recall but lacks in-context reasoning skill (23% indirect_any alone at $n=20$, condition C in Figure 9). Can the two be *composed* without inheriting LoRA’s damage? Our thesis: personal memory splits into *content* (per-user, trigger-keyable, low-cost) and *reasoning skill* (shared patterns, learnable from held-out users); storing each the right way beats every all-in-one baseline on every measure. Two hypotheses make this concrete:

- **H1 (naive combination):** per-user LoRA + per-user Engram (condition D) inherits LoRA’s indirect-reasoning damage; Engram’s local edit does not rescue the composition.
- **H2 (layered architecture):** one *shared* LoRA (cross-user reasoning skill) + per-user Engram (local content) splits memory correctly. The shared LoRA’s contamination is paid once and shared across all users, and the per-user Engram adds none on top ($\Delta\text{bpb}(F) = \Delta\text{bpb}(E)$ by Section 4’s locality property).

6.1 Design

Train a single shared LoRA on cross-user in-context-reasoning data: for each held-out training user (here, u020–u029), render each indirect QA as a completion-format sample “Facts: <fact1>. <fact2>. ... Q: <q> A: <gold>”, using only the facts in `required_fact_keys`. This teaches the LoRA the *pattern* “given facts in context, do the reasoning,” without memorising any specific user’s content. At inference, attach the shared LoRA + the test user’s Engram-row override and ask the indirect question without facts in the prompt: the Engram provides content (via trigger N-gram lookup), the shared LoRA provides the skill (Appendix Figure 25).

6.2 Six-condition head-to-head

We evaluate every per-user combination on $n=20$ test users ($n=10$ training users held out for the shared LoRA), measuring direct top-1/top-5 recall on the user’s facts (completion-format prompts), indirect top-1 and indirect_any on completion-format indirect probes (“Q: <q>\nA:”, greedy 16-token continuation), and `val_bpb` on a held-out 262 K-token ClimbMix shard (Figure 9).

6.3 Results: the layered design wins on every measure

The headline is the F-vs-B contrast (Figure 9). The layered design matches per-user LoRA on direct recall (100% vs. 99%) but answers indirect questions $7.4\times$ **more often** (44% vs. 6%), with $4.6\times$ **less damage to unrelated text** and **not a single user out of 20 made worse than the**

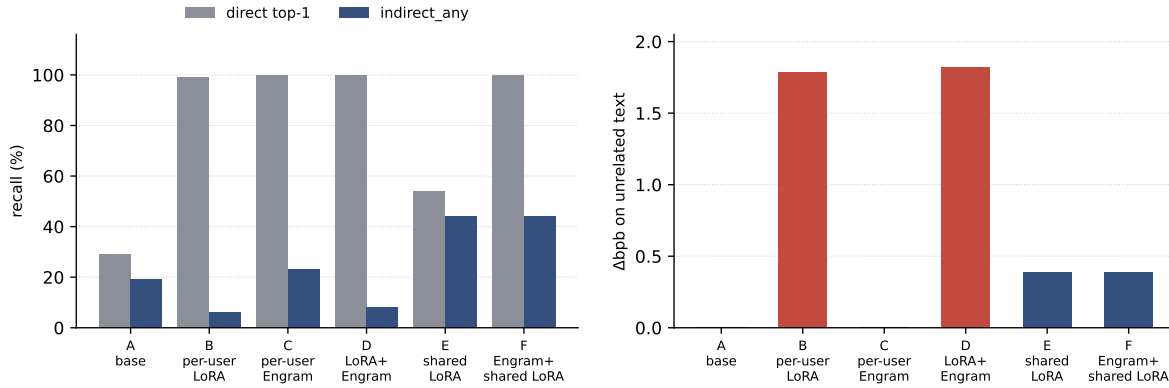


Figure 9. The six parametric conditions on $n=20$ test users (Mini-Engram-d20, seed S0). **Left:** F (Engram + shared LoRA) matches per-user LoRA’s direct recall (100% vs. 99%) while delivering $7.4\times$ its indirect_any (44% vs. 6%); neither the per-user Engram (C, no skill) nor the shared LoRA alone (E, never saw the user’s facts) suffices. **Right:** F adds *zero* contamination on top of the shared skill ($\Delta\text{bpb} + 0.386 = \text{E}$), while per-user LoRA (B) and the naive LoRA+Engram stack (D) pay $+1.78/+1.82$. RAG conditions (G–J) appear in Figure 21; 3 seeds in Appendix O (3-seed mean F 41%, B 7%).

untouched model (against 17 of 20 for the LoRA), all at 88 KB per user instead of 14.2 MB. The gap is stable across three seeds (paired-bootstrap 95% CI $[+31, +37]$ pp; Appendix O).

H1 holds: bolting an Engram onto a per-user LoRA does not rescue its reasoning—the LoRA’s damage dominates the combination. H2 holds on all three counts: adding the per-user Engram on top of the shared skill costs *zero* extra damage to unrelated text (its Δbpb equals the shared LoRA’s alone), keeps direct recall at 100%, and roughly doubles indirect accuracy over either the Engram alone or the untouched model. Neither piece works on its own—the Engram has the facts but no reasoning, the shared skill has reasoning but has never seen the user’s facts—and only together do they win.

6.4 Does RAG just close the gap instead?

The obvious objection to F is: why not just retrieve the facts and put them in the prompt? We test this two ways (conditions G–J in Figure 9; context-cost plot in Appendix H, Figure 21). On the Engram base itself, putting facts in the prompt sits *below* F—the base LM never learned to use an in-context fact list, and even handing it the exact right facts does not help, because the missing piece is the reasoning skill, not the facts. The fairer test feeds the retrieved facts to a real instruction-tuned model (Qwen2.5-3B-Instruct, about $2.5\times$ larger). Even there, F—with no retrieval and an empty prompt—lands within a few points of the larger model, and adding retrieval to our shared skill (J) passes it once context is free. The plain rule: no room in the prompt $\rightarrow$ use F; room to spare $\rightarrow$ add retrieval on top of the shared skill.

6.5 RAG-vs-Engram trade-offs sharpen as the KB grows

The experiments above used each user’s native 34 facts. Production fact counts are typically larger (10^2 – 10^3 : chat history, calendar, preferences, contacts). We augment each test user’s 34 facts with distractors sampled from the 30-user schema-family pool (u000–u029 minus the active user; 1008 facts) to reach $N \in \{34, 100, 200, 300, 500, 1000\}$. Indirect probes are unchanged; the retriever now discriminates the gold fact among N candidates. F is unaffected:

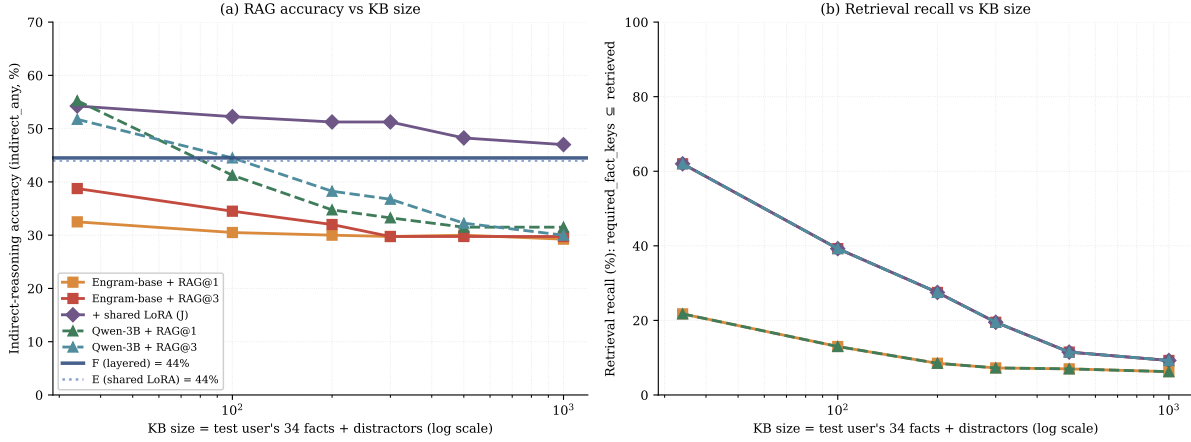


Figure 10. Indirect-reasoning accuracy and retrieval recall vs. KB size ($n=20$ users, 20 indirect probes each; KB = test user’s 34 facts + distractors sampled from the 30-user schema-family pool, log x -axis, $N \in \{34, 100, 200, 300, 500, 1000\}$). **(a)** F (layered, horizontal solid line at 44%) is invariant to KB size; the retrieval-based conditions degrade monotonically. **Naive RAG and Qwen-3B + RAG fall below F at KB ≥ 100 ; at KB=1000 the gap is 14 pp.** J (RAG top-3 + shared LoRA) is the most robust to KB growth (54% $\rightarrow$ 47%) because the shared reasoning skill compensates when retrieval misses, though its lead over F shrinks from 10 pp to 3 pp across the sweep. **(b)** Retrieval recall (fraction of probes where the retrieved set covers all gold required_fact_keys) drops by $7\times$ for top-3 (62% $\rightarrow$ 9%) between KB=34 and KB=1000, which is the mechanistic driver of (a).

the per-user Engram table holds only the test user’s facts, independent of population size.

Retrieval degrades quickly: as the candidate pool grows, top-3 recall on all-MiniLM-L6-v2 collapses several-fold purely from more candidates in the same embedding space, which drives the accuracy drop. Naive RAG, and even Qwen-3B + RAG, fall below F once the KB passes ~ 100 facts, while F holds flat—its per-user table never grows. The clean reading (Figure 10): *which method wins shifts with deployment scale*. At a few dozen facts, RAG on a larger model is the accuracy ceiling and F is the zero-context option; by a few hundred facts, F decisively wins.

Where each side’s failure comes from. The two limits are mechanistically distinct. Engram’s density ceiling (Section 5.2) comes from forward-pass interference among co-active rows inside one user’s table, and is bounded by per-user fact count. RAG’s retrieval ceiling comes from nearest-neighbour confusion across an ever-growing candidate pool, and is bounded by whatever pool the retriever sees (per-user, per-tenant, or per-corpus). For 10^2 – 10^3 facts/user, Engram density is the limiting factor on the parametric side; for 10^4 + facts in a shared corpus, retrieval is the limiting factor on the RAG side. The two costs scale on different axes and cross around $N \approx 100$.

7 Multi-tenant serving system

We implement EngramServer, a ~ 50 -line wrapper around an Engram-pretrained model. The base and global tables stay on the GPU; each user’s small override map lives in CPU memory. A request resolves the user, writes their rows, runs the forward pass, and restores the originals—no router, no custom kernel, no graph rewrite. Writing and restoring the rows cost about 2.3 ms each, and the forward pass dominates the rest (full breakdown in Appendix G).

It scales flat in the number of users. Appendix Figure 28 reports throughput, latency, storage, and recall across four deployment scales.

Per-request work does not depend on how many users share the server, so throughput and latency barely move as users triple (Appendix Figure 28); each extra user costs one small row-swap and no shared state. Two users are never in the table at the same time, so one cannot read another’s facts: cross-user leakage is zero by construction. (The only residual overlap is coincidence—two users with the same favourite spice—and it is small.) By contrast, serving per-user LoRAs needs custom CUDA kernels and batch-routing (S-LoRA [Sheng et al., 2024], Punica [Chen et al., 2024]); here the write is just an array assignment. Batched multi-user serving would add one gather per Engram layer, which we do not benchmark.

8 Discussion and limitations

8.1 When to use what

Which method to use turns on three things: *how many facts each user has*, *whether the queries need indirect reasoning*, and *whether context budget or per-user storage is the tighter constraint*. Table 2 gives the recommended method for each case.

Table 2. Cheat sheet for which method to use. Numbers cite this work unless marked external. “invariant” = the number does not degrade with KB / population size.

case	recommended method	per-user cost	headline metric	key citation
<10 facts/user, direct recall only	ICL / strong-retriever RAG	0 KB persistent	near-perfect recall	Sec. 5.4
10–1000 facts/user, indirect matters	layered F (Engram + shared LoRA)	88 KB , 0 ctx toks	44% indirect (invariant in KB)	Sec. 6–6.5
10–1000 facts/user, ctx budget free	J (RAG top-3 + shared LoRA)	88 KB + 44 ctx toks	54% indirect (KB=34)	Sec. 6.4
enterprise, 100% direct top-1, <10K users	per-user LoRA (S-LoRA serving)	14.2 MB	100% direct, indirect cost	Sec. 3
long-form / open-domain / shared corpus	RAG	0 KB persistent	wins LOCOMO open-domain +53%	Tab. 3

The most important row is the second: layered F is the right choice for the common case (10–1000 facts/user, indirect reasoning matters, many users), and decisively beats RAG with a $\sim 2.5\times$ larger instruction-tuned LM at $N \geq 100$. *Any* method that reaches 100% direct top-1 pays a price in indirect reasoning; the question is whether that price is baked in (a global LoRA), manageable (a local Engram, where the trigger N-gram is the contract), or set by retrieval (RAG, where the encoder must pick the gold fact out of N candidates).

8.2 Open limitations

Shared multi-hop reasoning gap. Both LoRA and Engram are surface-trigger keyed; neither composes facts across triggers (“if my doctor is Patel and Patel works at Globex, what is my doctor’s employer?”). User-as-Engram inherits this gap from per-user LoRA. A candidate fix is to mix recite-then-reason traces into Engram pretraining; we leave it to future work.

Within-user density ceiling. Joint OPT at 1000 facts reaches 35% top-1. The slot space is highly sparse ($< 1\%$ occupancy), so the constraint is forward-pass interference among co-active rows, not hash collisions. Per-user scoped tables (load only the relevant rows for the current query, via a small classifier) is a candidate future fix; another is the multi-fact-in-the-loss recipe change in Section 8.3.

Engram pretraining required. Our method assumes an Engram-pretrained base. We trained four ourselves at 178 M, 339 M, 625 M, and 1.22 B total parameters; production deployment depends on either DeepSeek releasing weights or pretraining your own (~10 h on a single Blackwell GPU at our 625 M scale; production-scale Engrams require correspondingly larger compute budgets).

8.3 Future work: training-recipe changes

Three changes to the training recipe target the limits we documented, and we expect more from them per unit of compute than from simply making the model bigger. (1) *Put many facts in the loss during training*, so the model learns to hold several at once. We tested a lighter version of this by finetuning d12@1280, and it helps at high density—but only by reaching the same recall in fewer optimisation steps, not by raising the ceiling (Appendix P.2, Figure 29). This pins the density ceiling on inference-time optimisation, not pretraining. (2) *Train with random per-user offsets to the hash*, which would let the gate read rows it never saw in training and bring back a per-user-salt option. (3) *Mix in “recall the fact, then reason” examples*, the most direct attack on the chaining gap.

A negative result worth recording. We tried improving the shared skill by training it on reasoning traces from a teacher model. It cut the contamination but also *lowered* accuracy (Appendix P). The reason is instructive: those traces assume the model first writes out a thinking step, while our test reads only the final answer. The recipe and the way you score it have to be designed together—you cannot mix and match them. Remaining layered follow-ups: a larger shared-skill training set, training the per-user table and the shared skill together, and starting from a chat-tuned base.

9 Conclusion

Personal memory is two jobs, not one: remembering a user’s specific facts, and the general skill of reasoning with them. The dominant recipe asks one per-user adapter to do both, and that is the root of the trouble: an adapter that must *hold* the facts cannot help also reshaping how the model thinks, so remembering and reasoning end up pulling on the same weights and the user pays for it in damaged knowledge and weaker reasoning. User as Engram separates the two—a user’s facts become a small, local edit, and the reasoning skill lives in one model everyone shares.

The sharpest open problem is the one our mechanism makes vivid: it can match its stored facts but cannot truly chain them. A memory that could combine its own traces, rather than only look them up, would close the last case where retrieval still wins on meaning; teaching the shared skill to keep learning as a user’s questions evolve is the natural companion step.

A century ago Richard Semon coined *engram* for the trace an experience etches into living tissue—a mark local enough that one memory does not erase the next. The brain pairs that sparse, local trace with a slow cortex that learns how to use it, and the pairing is what lets a person remember a new fact without forgetting how to think. Maya’s assistant needs the same discipline. When we stop asking one set of weights to be both the memory and the mind, and instead give each user a small private trace beside a skill everyone shares, personalization stops fighting the model it runs on. That, more than any single number, is what *User as Engram*

is for.

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References

- B. Li. User as code: Executable memory for personalized agents. arXiv:2606.16707, 2026.
- R. Semon. *The Mneme*. George Allen & Unwin, London, 1921. (English translation of *Die Mneme*, 1904; origin of the term “engram”).
- J. L. McClelland, B. L. McNaughton, and R. C. O’Reilly. Why there are complementary learning systems in the hippocampus and neocortex: Insights from the successes and failures of connectionist models of learning and memory. *Psychological Review*, 102(3):419–457, 1995.
- S. Tonegawa, X. Liu, S. Ramirez, and R. Redondo. Memory engram cells have come of age. *Neuron*, 87(5):918–931, 2015.
- D. Kumaran, D. Hassabis, and J. L. McClelland. What learning systems do intelligent agents need? Complementary learning systems theory updated. *Trends in Cognitive Sciences*, 20(7):512–534, 2016.
- X. Cheng, W. Zeng, D. Dai, Q. Chen, B. Wang, Z. Xie, K. Huang, X. Yu, Z. Hao, Y. Li, H. Zhang, H. Zhang, D. Zhao, and W. Liang. Conditional memory via scalable lookup: A new axis of sparsity for large language models. arXiv:2601.07372, January 2026.
- A. Vaswani, N. Shazeer, N. Parmar, J. Uszkoreit, L. Jones, A. N. Gomez, L. Kaiser, and I. Polosukhin. Attention is all you need. NeurIPS 2017.
- T. Brown et al. Language models are few-shot learners. NeurIPS 2020.
- A. Radford, J. Wu, R. Child, D. Luan, D. Amodei, and I. Sutskever. Language models are unsupervised multitask learners (GPT-2). OpenAI technical report, 2019.
- S. Min, X. Lyu, A. Holtzman, M. Artetxe, M. Lewis, H. Hajishirzi, and L. Zettlemoyer. Rethinking the role of demonstrations: What makes in-context learning work? EMNLP 2022.
- P. Lewis, E. Perez, A. Piktus, F. Petroni, V. Karpukhin, N. Goyal, H. Küttler, M. Lewis, W. Yih, T. Rocktäschel, S. Riedel, and D. Kiela. Retrieval-augmented generation for knowledge-intensive NLP tasks. NeurIPS 2020.
- Y. Gao et al. Retrieval-augmented generation for large language models: A survey. arXiv:2312.10997, 2023.
- N. Shazeer, A. Mirhoseini, K. Maziarz, A. Davis, Q. Le, G. Hinton, and J. Dean. Outrageously large neural networks: The sparsely-gated mixture-of-experts layer. ICLR 2017.

- D. Dai et al. DeepSeekMoE: Towards ultimate expert specialization in mixture-of-experts language models. arXiv:2401.06066, 2024.
- S. Mangrulkar et al. PEFT: State-of-the-art parameter-efficient fine-tuning methods. <https://github.com/huggingface/peft>, 2022.
- K. Jordan et al. Muon: An optimiser for hidden layers in neural networks. GitHub / blog post, 2024.
- C. Huang et al. LoRAHub: Efficient cross-task generalization via dynamic LoRA composition. arXiv:2307.13269, 2023.
- D. Wu et al. LongMemEval: Benchmarking chat assistants on long-term memory. arXiv 2024.
- A. Maharana et al. LOCOMO: Evaluating very long-term conversational memory of LLM agents. ACL 2024.
- M. Tavakoli, A. Salemi, C. Ye, M. Abdalla, H. Zamani, and J. R. Mitchell. Beyond a million tokens: Benchmarking and enhancing long-term memory in LLMs (BEAM). arXiv:2510.27246, 2025.
- B. Jiang, Y. Yuan, M. Shen, Z. Hao, Z. Xu, Z. Chen, Z. Liu, A. R. Vijjini, J. He, H. Yu, R. Pooven-dran, G. Wornell, L. Ungar, D. Roth, S. Chen, and C. J. Taylor. PersonaMem-v2: Towards personalized intelligence via learning implicit user personas and agentic memory. arXiv:2512.06688, 2025.
- A. Karpathy. nanochat: an experimental training harness for LLMs. <https://github.com/karpathy/nanochat>, 2026.
- E. Hu, Y. Shen, P. Wallis, Z. Allen-Zhu, Y. Li, S. Wang, L. Wang, and W. Chen. LoRA: Low-rank adaptation of large language models. ICLR 2022.
- N. Houlsby et al. Parameter-efficient transfer learning for NLP. ICML 2019.
- W. Su et al. Parametric retrieval-augmented generation (PRAG). arXiv:2501.15915, 2025.
- Z. Tan et al. DyPRAG: Dynamic parametric RAG. arXiv:2505.19386, 2025.
- J. Chen, H. Zhang, L. Pang, Y. Tong, H. Zhou, Y. Zhan, W. Lin, and Z. Zheng. Privacy-preserving reasoning with knowledge-distilled parametric retrieval-augmented generation (DistilledPRAG). arXiv:2509.01088, 2025.
- Z. Tan, Q. Liu, and M. Jiang. Democratizing large language models via personalized parameter-efficient fine-tuning (OPPU). EMNLP 2024 (arXiv:2402.04401).
- Y. Zhuang et al. HYDRA: Per-user adapters for personalised LLMs. arXiv 2024.
- M. Bini, O. Bohdal, U. Michieli, Z. Akata, M. Ozay, and T. Ceritli. MemLoRA: Distilling expert adapters for on-device memory systems. arXiv:2512.04763, 2025.
- R. Charakorn, E. Cetin, Y. Tang, and R. T. Lange. Text-to-LoRA: Instant transformer adaption. arXiv:2506.06105, 2025.

- Z. Tan et al. PER-PCS: Per-user post-hoc LoRA composition. arXiv 2024.
- Y. Sheng, S. Cao, D. Li, et al. S-LoRA: Serving thousands of concurrent LoRA adapters. arXiv:2311.03285, 2024.
- L. Chen, Z. Ye, Y. Wu, et al. Punica: Multi-tenant LoRA serving. MLSys 2024.
- G. Lample, A. Sablayrolles, M. Ranzato, L. Denoyer, and H. Jégou. Large memory layers with product keys. NeurIPS 2019.
- P. He. PEER: Mixture of one million experts. arXiv 2024.
- V. Berges, B. Oğuz, D. Haziza, W. Yih, L. Zettlemoyer, and G. Ghosh. Memory layers at scale. ICML 2025.
- Z. Huang, Q. Min, H. Huang, D. Zhu, Y. Zeng, R. Guo, and X. Zhou. Ultra-sparse memory network (Ultra-Mem). arXiv:2411.12364, 2024 (ICLR 2025).
- J. Huang et al. OverEncoding: hashed N-gram embeddings via averaging. 2025.
- L. Yu et al. SCONE: scalable contextual N-gram embeddings. 2025.
- A. Pagnoni, R. Pasunuru, P. Rodriguez, et al. BLT: byte latent transformer with hashed N-gram embeddings. arXiv:2412.09871, 2025.
- A. Liu et al. SuperBPE: word-level BPE for compositional patterns. 2025.
- CivitAI Community. LoRA stacking patterns for Stable Diffusion. <https://civitai.com/>, 2024.
- K. Meng, D. Bau, A. Andonian, and Y. Belinkov. Locating and editing factual associations in GPT (ROME). NeurIPS 2022.
- K. Meng et al. MEMIT: Mass-editing memory in a transformer. ICLR 2023.
- R. Cohen et al. Evaluating the ripple effects of knowledge editing in language models (MQuAKE). 2023.
- R. Cohen et al. RippleEdits: A benchmark for ripple effects of model editing. 2024.
- K. Meng et al. CounterFact: a counterfactual editing benchmark. 2022.
- D. Wu, J.-C. Gu, K.-W. Chang, and N. Peng. Self-routing RAG: Binding selective retrieval with knowledge verbalization. arXiv:2504.01018, 2025.
- Z. Sun et al. Recitation-augmented language models. ICLR 2023.
- A. Yang, B. Yang, B. Zhang, et al. Qwen2.5 technical report. arXiv:2412.15115, 2025.
- A. Yang, A. Li, B. Yang, et al. Qwen3 technical report. arXiv:2505.09388, 2025.
- A. Grattafiori, A. Dubey, A. Jauhri, et al. The Llama 3 herd of models. arXiv:2407.21783, 2024.
- A. Q. Jiang, A. Sablayrolles, A. Mensch, et al. Mistral 7B. arXiv:2310.06825, 2023.

- DeepSeek-AI. DeepSeek-V3 technical report. arXiv:2412.19437, 2024.
- N. Reimers and I. Gurevych. Sentence-BERT: Sentence embeddings using Siamese BERT-networks. EMNLP-IJCNLP 2019.
- W. Wang, F. Wei, L. Dong, H. Bao, N. Yang, and M. Zhou. MiniLM: Deep self-attention distillation for task-agnostic compression of pre-trained transformers. NeurIPS 2020.
- L. Zheng, W.-L. Chiang, Y. Sheng, et al. Judging LLM-as-a-judge with MT-Bench and Chatbot Arena. NeurIPS 2023 Datasets and Benchmarks. arXiv:2306.05685.
- C. Packer, S. Wooders, K. Lin, et al. MemGPT: Towards LLMs as operating systems. arXiv:2310.08560, 2023.
- P. Chhikara, D. Khant, S. Aryan, T. Singh, and D. Yadav. Mem0: Building production-ready AI agents with scalable long-term memory. arXiv:2504.19413, 2025.
- W. Xu, Z. Liang, K. Mei, H. Gao, J. Tan, and Y. Zhang. A-MEM: Agentic memory for LLM agents. arXiv:2502.12110, 2025.
- P. Rasmussen, P. Paliychuk, T. Beauvais, J. Ryan, and D. Chalef. Zep: A temporal knowledge graph architecture for agent memory. arXiv:2501.13956, 2025.
- Z. Li, S. Song, H. Wang, et al. MemOS: An operating system for memory-augmented generation in large language models. arXiv:2505.22101, 2025.
- S. Wang, E. Yu, O. Love, T. Zhang, T. Wong, S. Scargall, and C. Fan. MemMachine: A ground-truth-preserving memory system for personalized AI agents. arXiv:2604.04853, 2026.
- C. Hu, X. Gao, Z. Zhou, et al. EverMemOS: A self-organizing memory operating system for structured long-horizon reasoning. arXiv:2601.02163, 2026.
- D. Patel and S. Patel. ENGRAM: Effective, lightweight memory orchestration for conversational agents. arXiv:2511.12960, 2025.
- S. Yan, X. Yang, Z. Huang, et al. Memory-R1: Enhancing large language model agents to manage and utilize memories via reinforcement learning. arXiv:2508.19828, 2025.
- Y. Yu, L. Yao, Y. Xie, et al. Agentic memory: Learning unified long-term and short-term memory management for LLM agents. arXiv:2601.01885, 2026.
- Y. Wang, R. Takanobu, Z. Liang, et al. Mem- α : Learning memory construction via reinforcement learning. arXiv:2509.25911, 2025.
- Z. Zhang, X. Bo, C. Ma, et al. A survey on the memory mechanism of large language model based agents. arXiv:2404.13501, 2024.
- Y. Wu, S. Liang, C. Zhang, et al. From human memory to AI memory: A survey on memory mechanisms in the era of LLMs. arXiv:2504.15965, 2025.
- P. Du. Memory for autonomous LLM agents: Mechanisms, evaluation, and emerging frontiers. arXiv:2603.07670, 2026.

- N. Pollertlam and W. Kornsuwannawit. Beyond the context window: A cost-performance analysis of fact-based memory vs. long-context LLMs for persistent agents. arXiv:2603.04814, 2026.
- A. Salemi, S. Mysore, M. Bendersky, and H. Zamani. LaMP: When large language models meet personalization. ACL 2024. arXiv:2304.11406.
- Z. Zhang, R. A. Rossi, B. Kveton, et al. Personalization of large language models: A survey. arXiv:2411.00027, 2024.
- J. Liu, Z. Qiu, Z. Li, et al. A survey of personalized large language models: Progress and future directions. arXiv:2502.11528, 2025.
- Y. Xu, Q. Chen, Z. Ma, et al. Toward personalized LLM-powered agents: Foundations, evaluation, and future directions. arXiv:2602.22680, 2026.
- E. Mitchell, C. Lin, A. Bosselut, C. Finn, and C. D. Manning. Fast model editing at scale. ICLR 2022.
- E. Mitchell, C. Lin, A. Bosselut, C. D. Manning, and C. Finn. Memory-based model editing at scale. ICML 2022.
- D. Dai, L. Dong, Y. Hao, Z. Sui, B. Chang, and F. Wei. Knowledge neurons in pretrained transformers. ACL 2022.
- P. Wang, Z. Li, N. Zhang, et al. WISE: Rethinking the knowledge memory for lifelong model editing of large language models. NeurIPS 2024.
- Y. Yao, P. Wang, B. Tian, et al. Editing large language models: Problems, methods, and opportunities. EMNLP 2023.
- M. McCloskey and N. J. Cohen. Catastrophic interference in connectionist networks: The sequential learning problem. *Psychology of Learning and Motivation*, 24:109–165, 1989.
- J. Kirkpatrick, R. Pascanu, N. Rabinowitz, et al. Overcoming catastrophic forgetting in neural networks. *PNAS*, 114(13):3521–3526, 2017.
- H. Shi, Z. Xu, H. Wang, et al. Continual learning of large language models: A comprehensive survey. arXiv:2404.16789, 2024.
- K. Guu, K. Lee, Z. Tung, P. Pasupat, and M.-W. Chang. REALM: Retrieval-augmented language model pre-training. ICML 2020.
- U. Khandelwal, O. Levy, D. Jurafsky, L. Zettlemoyer, and M. Lewis. Generalization through memorization: Nearest neighbor language models. ICLR 2020.
- S. Borgeaud, A. Mensch, J. Hoffmann, et al. Improving language models by retrieving from trillions of tokens. ICML 2022.
- G. Izacard, P. Lewis, M. Lomeli, et al. Atlas: Few-shot learning with retrieval augmented language models. *JMLR*, 24(251):1–43, 2023.

- A. Asai, Z. Wu, Y. Wang, A. Sil, and H. Hajishirzi. Self-RAG: Learning to retrieve, generate, and critique through self-reflection. ICLR 2024.
- M. Geva, R. Schuster, J. Berant, and O. Levy. Transformer feed-forward layers are key-value memories. EMNLP 2021.
- X. L. Li and P. Liang. Prefix-tuning: Optimizing continuous prompts for generation. ACL 2021.
- Q. Zhang, M. Chen, A. Bukharin, et al. AdaLoRA: Adaptive budget allocation for parameter-efficient fine-tuning. ICLR 2023.
- T. Detrmers, A. Pagnoni, A. Holtzman, and L. Zettlemoyer. QLoRA: Efficient finetuning of quantized LLMs. NeurIPS 2023.

Appendix A: LOCOMO full results

Figure 11 summarizes the category breakdown; Table 3 gives the underlying per-cell token-F1 for every system, scale, and category.

Table 3. LOCOMO category $\times$ model token-F1. Engram Joint OPT vs. the best retrieval baseline (MEM0_LIKE or MEMMACHINE_LIKE) per cell. The Engram win is category-dependent (Figure 11).

category	answering LM	MEM0	MEMMACH	Engram J-OPT	lead vs. best retr.
single-hop	d12 v2	0.131	0.116	0.169	+29%
	d12@1280	0.160	0.152	0.176	+10%
	d20	0.161	0.169	0.233	+38%
multi-hop	d12 v2	0.092	0.076	0.243	+165%
	d12@1280	0.115	0.104	0.253	+120%
	d20	0.102	0.108	0.299	+178%
reasoning	d12 v2	0.117	0.113	0.183	+56%
	d12@1280	0.123	0.136	0.203	+49%
	d20	0.135	0.168	0.266	+58%
open-domain	d12 v2	0.247	0.209	0.122	−51%
	d12@1280	0.341	0.327	0.146	−57%
	d20	0.314	0.338	0.159	−53%

The two diagnostic figures below support the LLM-judge discussion of Section 5.5: Figure 12 shows that token-F1 over-credits Engram relative to an LLM judge, and Figure 13 shows that multi-token Joint-OPT closes the resulting judge gap.

Appendix B: Mechanistic analysis

This appendix consolidates the mechanistic evidence that is referenced piecewise in the main text (locality in Section 4.3, the multi-hop decomposition in Section 5.6, the read/write probe in Appendix I) into a single account of *how* a per-user fact becomes a retrievable, leak-free parametric edit. The picture is deliberately simple: User-as-Engram is not an opaque learned circuit

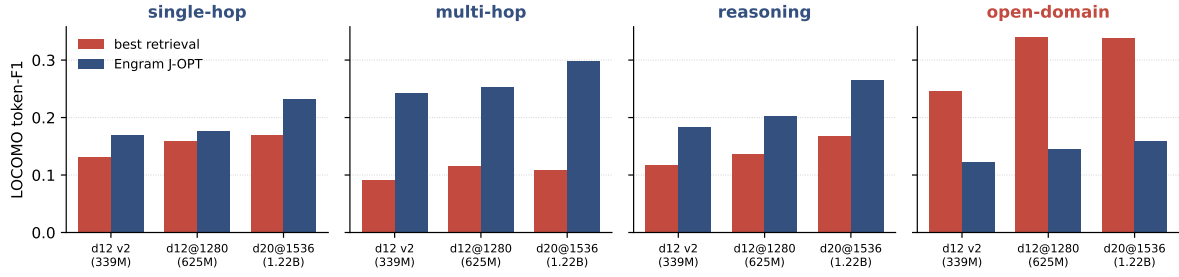


Figure 11. Locomo category breakdown (token-F1), Engram Joint OPT (blue) vs. the best retrieval baseline (red; max of MEM0_LIKE and MEMMACHINE_LIKE) across three dense scales. Engram wins single-hop, multi-hop, and reasoning at every scale—the per-fact (question, answer) loss encodes the question→answer map that retrieval must chain across evidence sentences—but loses open-domain, where the answer is a verbatim span of an unseen evidence sentence that a single-token bias cannot reconstruct.

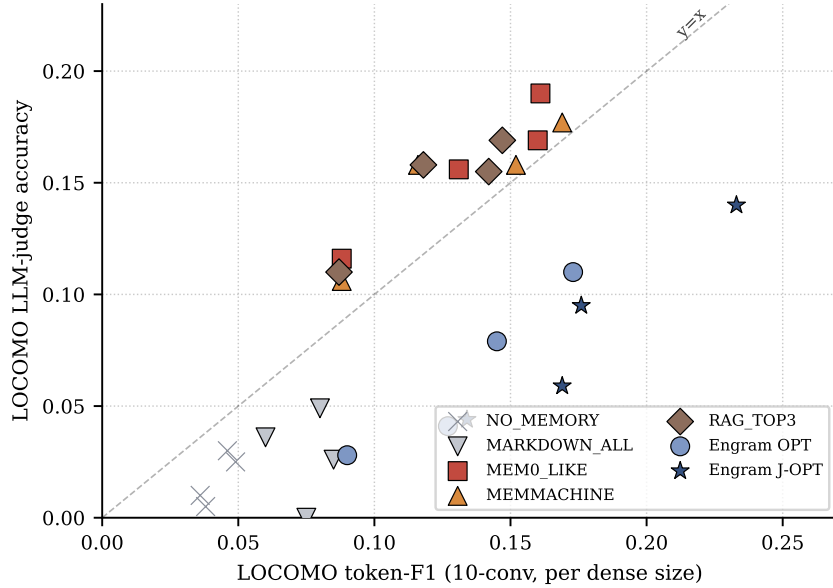


Figure 12. Locomo token-F1 vs. LLM-judge accuracy across all system × dense-size cells. Retrieval baselines (red/orange) sit near the $y=x$ line: their token-F1 and LLM-judge scores roughly agree. User-as-Engram OPT and Joint OPT (blue/cyan) sit systematically *above* the line, indicating that token-F1 over-credits Engram. The asymmetry is the metric mismatch documented in Section 5.5.

but a **content-addressable memory** whose every step is either deterministic (hashing, gating) or directly measurable (the value the row writes, the gate it opens, the locality of its effect). We trace the path of one inserted fact through five observable stages, verifying each on the trained Mini-Engrams (d12@1280 and d20, 16 canonical facts; probes in `scripts/mech_glassbox.py`, `mech_lora_vs_engram.py`, `mech_depth.py`). All numbers below are on the trained models unless stated.

(1) Addressing: a fact’s trigger N-gram is a hash key. A fact is written at the row(s) that its trigger suffix N-gram hashes to, via the deterministic multiplicative-XOR hash of the Engram architecture [Cheng et al., 2026]. The address is a pure function of surface tokens—no learning, no per-user state—which is why two users’ edits never interfere unless their *distinguishing*

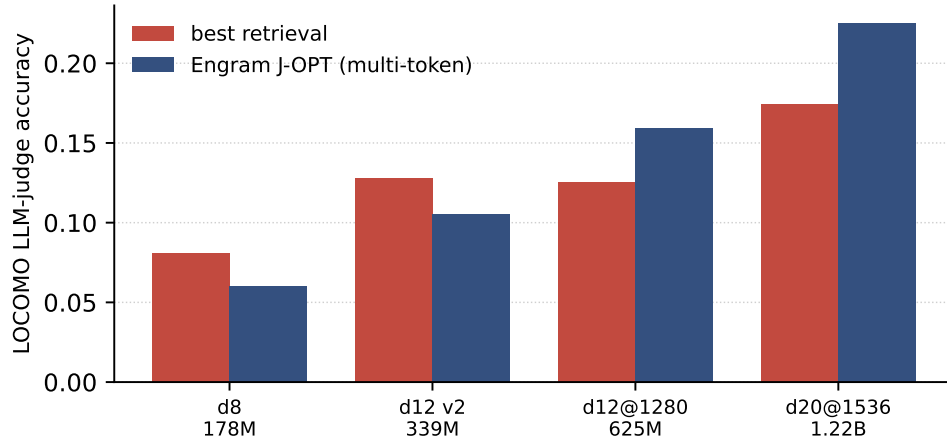


Figure 13. LOCOMO single-hop under an LLM judge (Qwen2.5-14B), matched multi-token pipeline. First-token OPT lets token-F1 over-credit Engram for a correct first token over a noisy continuation (Figure 12); training the *full* answer with multi-token Joint-OPT fixes this and overtakes the best retrieval baseline from d12@1280 upward (+27% at 625 M, +29% at 1.22 B). Below that cross-over the dense backbone is too small for the continuation to fall into place after the Engram-anchored prefix.

tokens collide. The T1 collision audit (Appendix I) measures this directly: across 100 synthetic users $\times$ 30 facts, pairwise overlap on user-distinguishing key tokens (Patel, Portland, 1991) is just 4.5%, and a 30-fact user occupies 0.027% of the demo table. Addressing is sparse and content-keyed, so capacity is never the limiting factor (Section P.3).

(2) The write opens its own gate and injects exactly its value path. A row reaches the residual stream only through the gated value $\alpha_t W_{ve}$, so a write has to do two things, and we can watch both. *It opens its own gate.* Because the gate’s key is W_{ke} , writing the row raises the trigger-position gate from $\alpha \approx 0.015$ (an untrained synthetic trigger barely fires) to $\alpha \approx 0.99$ under the deployed OPT strategy (0.59 under closed-form UNEMBED_P), while every non-trigger position stays at $\alpha \approx 0.03$ –0.04 (identical on d12@1280 and d20). *It injects its value path.* The residual change the deployed OPT row induces at the trigger has cosine **0.998** (d12@1280) / **0.999** (d20) with the analytic W_{ve} projection—the gate and short-convolution scale the injection but do not redirect it. This reproduces the random-init read/write probe (cosine 0.998, Appendix I) on the trained model, for the strategy we deploy. (The small-magnitude closed-form marker is dominated by the convolution nonlinearity, so its *direction* is noisier; OPT drives the row norm up, $99 \rightarrow 132$, until the value path dominates.) The trained model corroborates this from the behavioural side via the sensitivity asymmetry (Section 5.1): ablating the Engram pathway (engram_suppress) degrades *factual* top-5 recall (93.3% retained) while *reading* comprehension is untouched (100% retained)—a 6.7 pp asymmetry that localises factual content to the Engram pathway.

(2b) What the row encodes vs. what it optimises. A subtler reading concerns the row’s relation to the gold token. The closed-form UNEMBED_P row points its value path straight at the gold token’s unembedding (cosine **0.59**–**0.65** to $\text{lm_head}[y]$): “the row stores the gold value” is then literally true. Gradient OPT and Joint-OPT trade some of that direct alignment (cosine drops to 0.16–0.24) for higher recall, shaping the *full* next-token head rather than only the gold

unembedding direction. The interpretability is therefore in the exact value-path injection (stage 2), not in the row equalling a single unembedding vector—the deployed row is a value the network reads cleanly, optimised against the true objective.

(3) Locality is exact—and it comes from addressing, not from the gate. It is tempting to attribute locality to the gate (“it fires only on the trigger”), but the gate measurement above shows the gate does *not* single out an arbitrary synthetic trigger before the write. Locality has a simpler source: we write only the trigger’s rows, and the trigger’s suffix N -gram is unique in the sequence, so every other position reads unchanged rows and its output is bit-identical. We verify the consequence not on one fact but across *all 16 canonical facts and both insertion strategies* (`mech_glassbox.py`): the maximum residual change over every non-trigger position and every layer is **0.000**, and the change before the Engram layer is **0.000**—exact, through every subsequent attention/MLP layer (the trigger position itself moves by order 1). Figure 5 contrasts this with a per-user LoRA fit to the *same* fact, whose effect is nonzero at every position and every layer and which perturbs unrelated text by mean 107. Because the edit is *addressed*, a per-user write contaminates no unrelated forward pass ($\Delta\text{bpb} +0.00005$ vs. a global LoRA’s $+1.78$; Section 3), and cross-user leakage is zero by design (Section 4).

(4) Depth: the edit lands where the model has already “deepened”. Engram rows are read at a late Engram layer, after the network has effectively committed to a next-token distribution. The LogitLens trace (Figure 14) shows Mini-Engram-d8 converging to its output distribution *faster* than the base—layer-3 KL is 3.66 lower—reproducing Cheng et al.’s “effective deepening” at our scale. A row therefore overrides a near-final prediction at exactly the position where it is decoded, rather than steering an early, still-malleable representation (the place where a global LoRA change does its damage). We make this dependence *causal* rather than merely correlational by re-running the insertion at the model’s *early* Engram layer instead of the late one (layer 2 vs. layer 11 on d20, layer 2 vs. 7 on d12@1280; identical OPT-15 budget, `mech_depth.py`): top-1 recall collapses from **1.00 at the late layer to 0.25 (d20) / 0.31 (d12@1280) at the early layer**, because an early injection must survive the rest of the transformer stack and largely washes out. The depth at which the Engram reads is not incidental—it is where a value-path edit can override the prediction.

(5) Limit: the gate is a surface- N -gram hash, so it cannot compose across triggers. The same mechanism that makes the edit local also bounds what it can do. The multi-hop decomposition (Section 5.6, Figure 8) is the cleanest mechanistic readout in the paper: on a balanced 63-pair corpus, chained queries succeed **91%** of the time when the query’s suffix N -gram *overlaps* the second fact’s trigger, but only **13%** ($\lesssim 10\%$ after filtering first-fact-wins coincidences) when true composition is required—even though per-fact direct recall is 99.2% in both subsets. The near-bimodal split is exactly what a surface-keyed lookup predicts: the gate fires on a matching trigger and returns the stored token, or it does not fire and the gold token is far from the top. There is no chaining circuit to find because, by design, there is none.

Synthesis. The five stages compose into one sentence: *User-as-Engram hashes a fact’s surface trigger to a sparse row whose write both opens its own gate ($\alpha 0.02 \rightarrow 0.99$) and injects exactly its value path (cosine 0.999 to the predicted projection), at a late, already-deepened layer (early insertion recall*

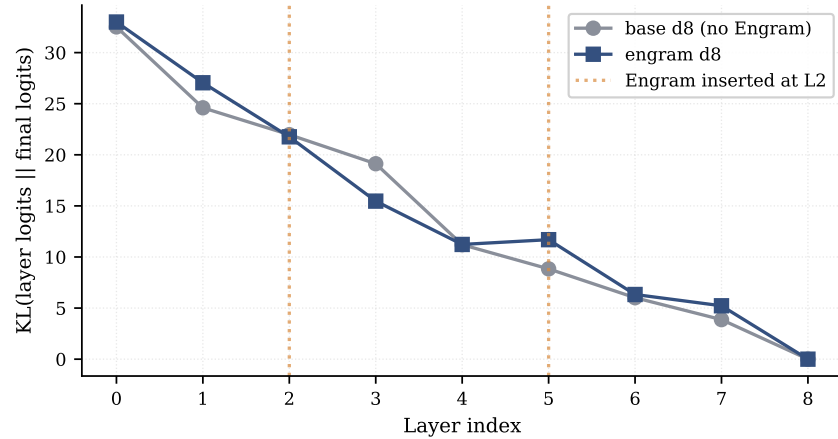


Figure 14. LogitLens KL by layer (Mini-Engram-d8 vs. base-d8). Engram converges faster at layer 3 (–3.66 KL gap), confirming Cheng et al.’s effective-deepening claim at our scale (stage 4 of the mechanism above).

collapses 1.00→0.25), with the change exactly 0.000 at every other position—and therefore retrieves single facts with zero contamination but cannot compose across facts. This is why the method cleanly separates *content* (the rows) from *reasoning skill* (composition, which must come from the shared LoRA in the layered design of Section 6)—the content/skill split the paper proposes is not an engineering convenience but a direct consequence of the mechanism.

Appendix C: Pretraining curves

Figure 15 shows the Mini-Engram pretraining runs that every later insertion experiment builds on. Adding the Engram table is not a tax on language modelling: at iso-FLOPs the d8 Engram model reaches marginally lower validation bits-per-byte than the d8 baseline (0.924 vs. 0.929 at step 4 000), reproducing Cheng et al.’s effective-deepening finding at our much smaller scale, and the larger d12 Engram model—more backbone, longer training—is substantially better (0.849 bpb). These checkpoints are the base models for the row-insertion results in the body.

Appendix D: Insertion strategy comparison

Figure 16 compares five ways of writing a fact into Engram rows at the hardest density we test (100 facts/user, Mini-Engram-d12). A random marker (RANDOM, a control) and the token input embedding (WTE) never recover the fact (0% top-1), confirming the recall signal is not an artefact of the metric. The closed-form pseudo-inverse marker (UNEMBED_P) reaches only 6%/18% top-1/top-5: it orients the row correctly but the gate suppresses its magnitude. Fifteen steps of per-row gradient descent (OPT-15) lift this to 38%/44%, and Joint OPT—optimising all touched rows together against the user’s full fact set—reaches 68%/96%, recovering most of what a per-user LoRA achieves at this density (Figure 26) at $161\times$ less storage. The earlier single-fact-at-a-time comparison on the original 16-fact USER+ORG benchmark (Mini-Engram-d8) is reported in Appendix K.

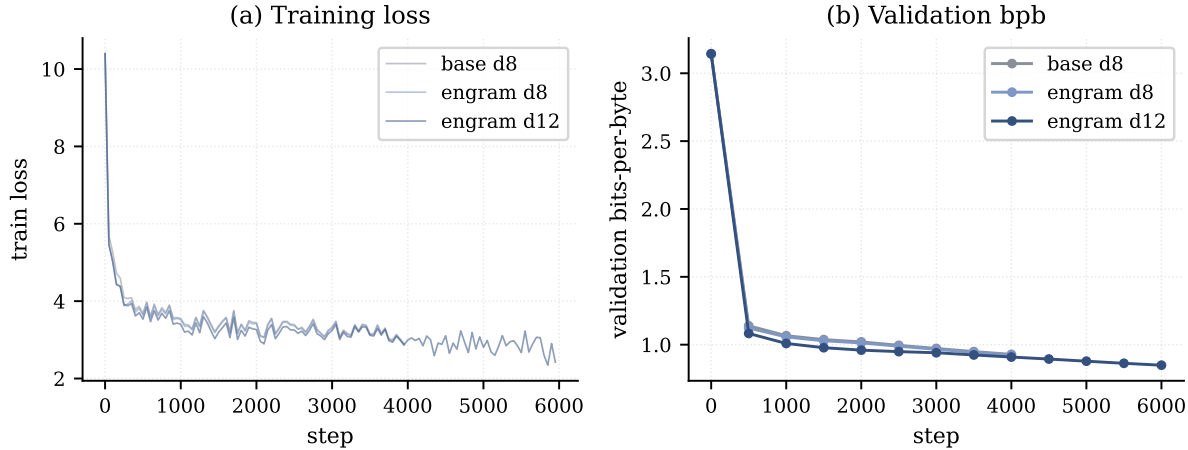


Figure 15. Mini-Engram pretraining curves. Engram d8 reaches lower validation bpb than base d8 at iso-FLOPs (matching the paper’s finding at much smaller scale); engram d12 is substantially better thanks to the larger backbone and longer training.

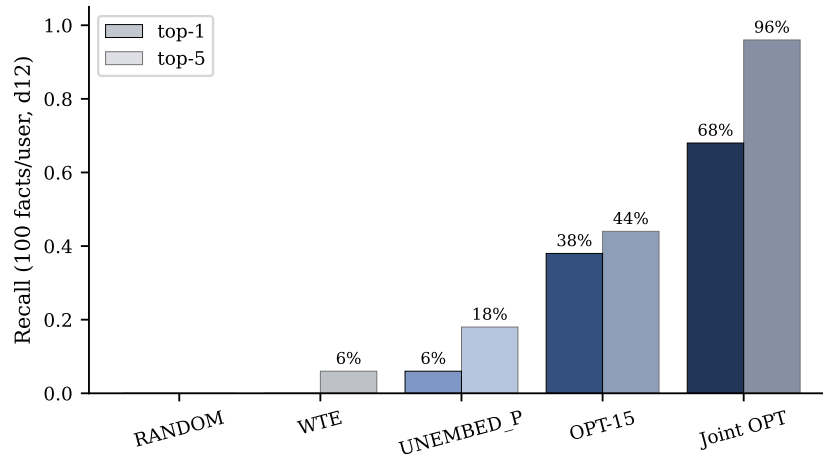


Figure 16. Top-1 and top-5 recall at 100 facts/user, Mini-Engram-d12. RANDOM (control) confirms our signal isn’t noise; UNEMBED_P alone is too weak; OPT-15 helps; Joint OPT closes most of the gap to full LoRA fine-tuning at $161\times$ less storage.

Appendix E: Paraphrase generalisation

Figure 17 tests whether a fact written at one trigger phrasing answers the same question asked differently, across five held-out paraphrases per fact. Single-trigger insertion (write once, at the canonical phrasing) averages 50% top-1: 4/5 for the “doctor” fact, 3/5 for “Globex hours”, 2/5 for “Stark HQ”, and 1/5 for “spice”. Generalisation tracks suffix N -gram overlap—paraphrases that end in the same tokens as the trained trigger transfer for free, while divergent surface forms (e.g. “I love the spice” vs. “my favorite spice is”) do not. Multi-trigger insertion (write the fact at all five phrasings, at $5\times$ the per-fact OPT cost) drives every paraphrase to top-1. The full per-paraphrase rank data are tabulated in Appendix L.

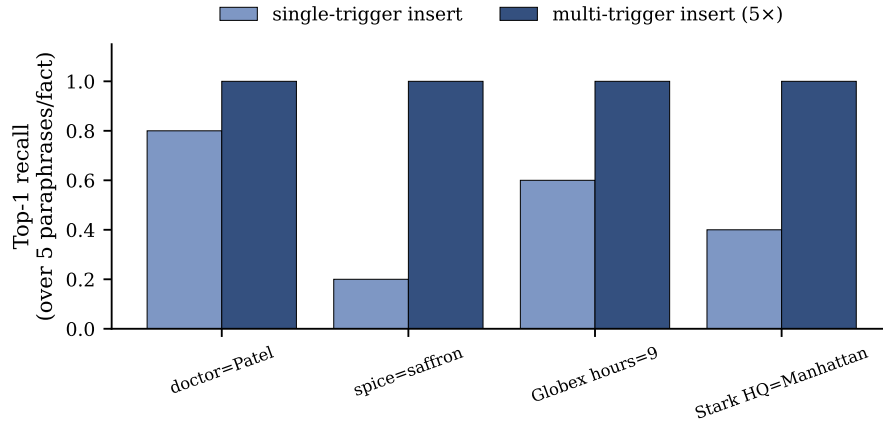


Figure 17. Paraphrase generalisation across 4 facts $\times$ 5 paraphrases each. Single-trigger insertion gets 50% top-1 for free (suffix N-gram overlap); multi-trigger insertion (insert at all 5 paraphrases) gets 100% at 5 $\times$ the per-fact OPT cost.

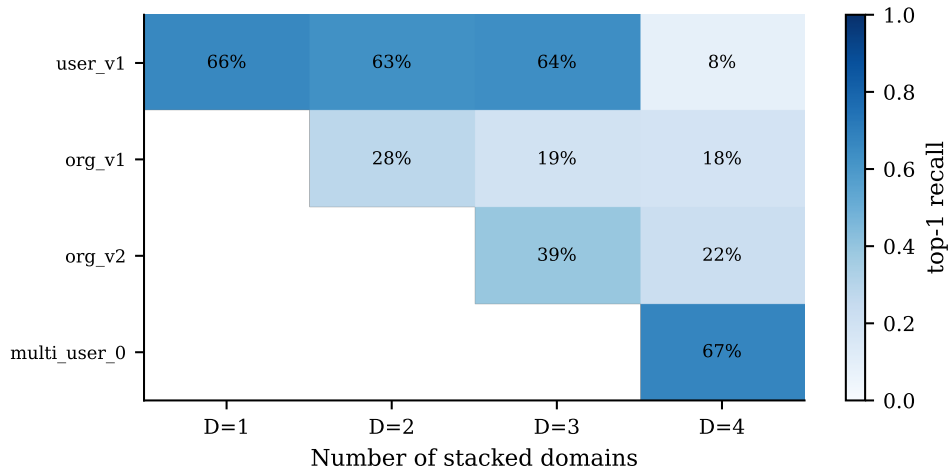


Figure 18. Multi-domain additive composition heatmap. Each cell shows per-domain top-1 recall when D domains are stacked. Heavy degradation when domains share trigger templates (high address overlap). Disjoint-template domains (corp + user, demonstrated in main text) compose without loss.

Appendix F: Multi-domain composition

Figure 18 stacks D fact domains into one Engram table and reports per-domain top-1 as D grows. The user domain (user_v1) is unaffected by adding disjoint-template corporate domains: it holds 66% $\rightarrow$ 63% $\rightarrow$ 64% as org_v1 and org_v2 are layered in ($D=1$ to 3). The collapse appears at $D=4$, when a second *user-template* domain (multi_user_0) is added: because it reuses the same trigger templates, the two user domains collide in the address hash and user_v1 falls to 8% while the newcomer reads 67%. Domains that share templates interfere (the corporate domains also degrade each other, org_v1 28% $\rightarrow$ 18%), whereas template-disjoint domains compose without loss—consistent with the corp+user composition measured directly in Section 5.3.

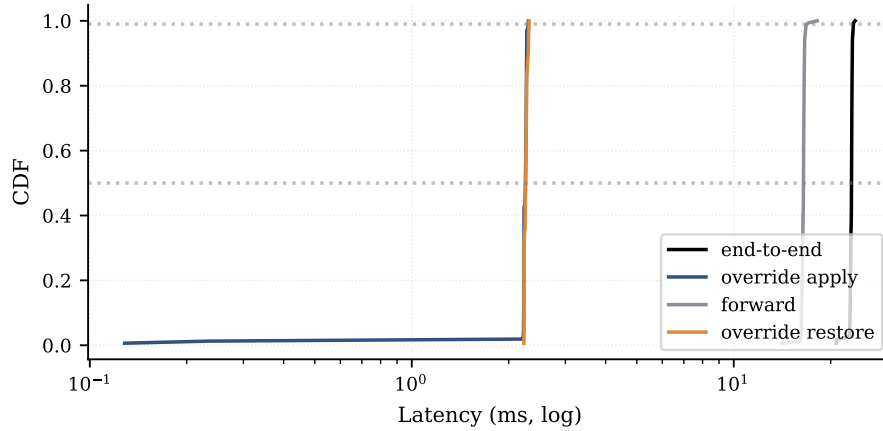


Figure 19. Per-request latency CDF on Mini-Engram-d12 (30 users $\times$ 50 facts $\times$ 600 requests). Override apply (blue) and restore (orange) are both sub-3 ms at p99; the forward pass (grey) dominates total latency (black).

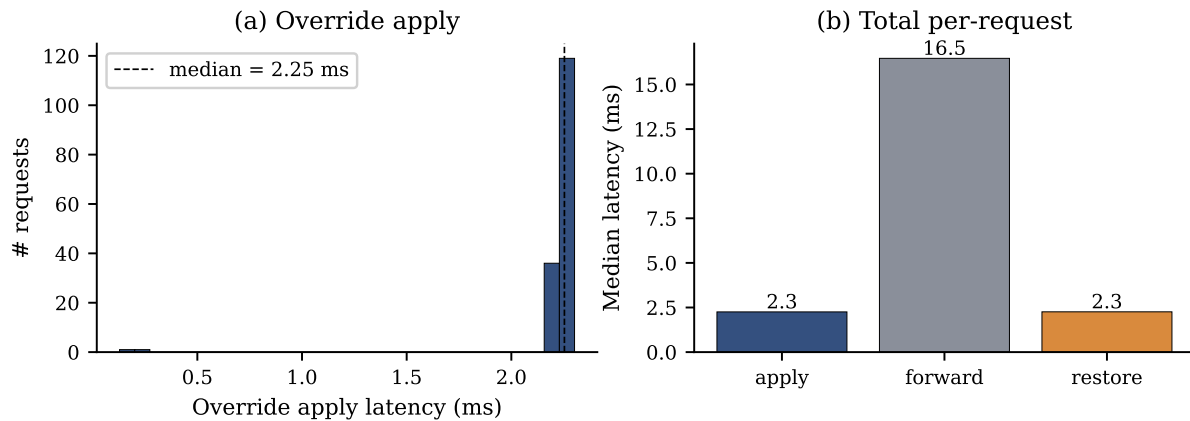


Figure 20. Per-request latency breakdown, Mini-Engram-d12 (30 users $\times$ 50 facts $\times$ 600 requests). **(a)** Override-apply latency: median 2.2 ms. **(b)** Median per-request latency by component: apply (2.3 ms) + forward (16.5 ms) + restore (2.3 ms); the frozen forward pass dominates, and the override apply+restore overhead (~ 4.6 ms) is under 20% of the 23.2 ms end-to-end median.

Appendix G: Serving latency CDF

Figure 19 shows the per-request latency distribution under multi-tenant serving (Mini-Engram-d12, 30 users $\times$ 50 facts, 600 requests). Applying a user’s override map and restoring the base table afterwards are both cheap and tightly bounded—median 2.2 ms, p99 2.3 ms each—so the swap adds no latency tail. The forward pass dominates: a median of 16.5 ms out of a 23.2 ms end-to-end total (p99 27.8 ms), so per-user memory adds under 20% overhead on top of a request the model would run anyway. The corresponding throughput across serving configurations is reported in Figure 28.

Appendix H: Qwen-3B + RAG details

The Qwen-3B + RAG measurements in Section 6.4 use Qwen/Qwen2.5-3B-Instruct in bf16 with greedy decoding (≤ 32 new tokens). The chat-template prompt is:

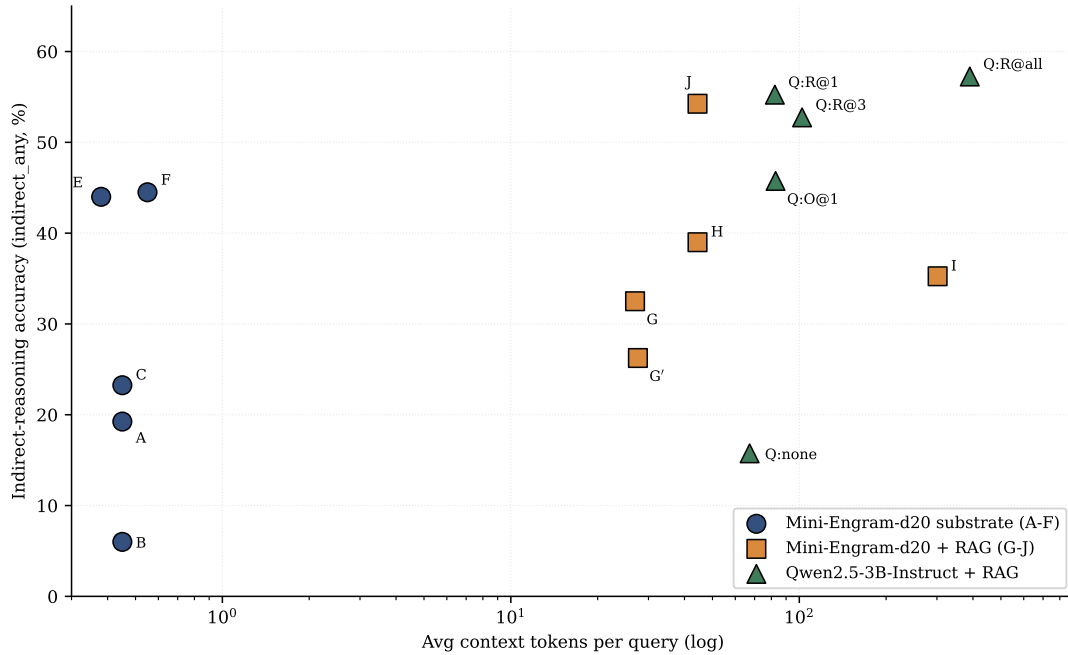


Figure 21. Indirect-reasoning accuracy (indirect_any) vs. avg context tokens per query on the 20-user split. F (layered) and E (shared LoRA only) anchor the 0-context end; J (RAG top-3 + shared LoRA) lifts accuracy to 54% at 44 context tokens; Qwen-3B + RAG reaches 55–57% at 82–390 tokens on a $\sim 2\times$ larger backbone. Naive RAG on the same Mini-Engram-d20 base (G/H/I) sits *below* F: putting facts in context is not the same as having the reasoning skill.

[SYSTEM]

You are answering personal-memory questions about the user. Use only the facts provided. Answer with the shortest possible span (typically 1-3 words). Do not explain or add any extra text.

[USER]

Facts about the user:

- <retrieved fact 1>.
- <retrieved fact 2>.
- ...

Question: <indirect question from indirect_qa>

Answer:

Retrieval uses the same sentence-transformers/all-MiniLM-L6-v2 encoder as the RAG/MEM0/MEMMACHINE baselines of Section 5.4, with the per-user fact set as the index. For each indirect probe the question text is embedded and cosine-similarity top- k retrieval returns the facts to put in the context block. The same 20-user/20-probe split as Figure 9 is evaluated.

Two answer-match criteria are reported: *indirect_top1* requires the first generated word to equal (case-insensitive) the first word of the gold answer; *indirect_any* requires the gold span to occur anywhere in the generation. This matches the substring matcher used for F.

Why oracle-top-1 underperforms RAG top-1. A non-obvious result in the Qwen-3B + RAG experiment is that oracle top-1 retrieval (using `required_fact_keys` to fetch the ground-truth

fact) gives 46% indirect_any, *below* RAG top-1’s 55%. Inspecting the data, retrieval often returns facts that are *adjacent* to the required key in semantic space and that happen to also contain the answer surface (e.g. retrieving the “birth-year” fact when asked about “current age” covers both the year and the implicit age). RAG with $k = 1$ thus over-samples the information needed for indirect reasoning, while strict oracle retrieval leaves the LM to derive the answer from one single-attribute fact—which Qwen-3B often refuses to do under our “shortest span” system prompt.

Appendix I: Preliminary architectural validation

Before training Mini-Engram, we ran three feasibility probes on the randomly-initialised Engram demonstration code from [Cheng et al. \[2026\]](#) to verify that the architecture admits surgical insertion at all. These results motivated the larger experiments in the body but are not central to the main claims; we report them for completeness.

T1 collision audit (100 synthetic users $\times$ 30 facts). Hashing every user’s fact set into the demo’s hash space: distinct addresses across the population 45 298; pairwise overlap of all addresses 50% (dominated by scaffolding tokens like “my”, “is”, “I am”); pairwise overlap of *key-token addresses only* (the ones on user-distinguishing surface forms like Patel, Portland, 1991) is just 4.5%. For a 30-fact user the slot-occupancy fraction is 0.027%; the 1.6 M-slot demo table is 99.97% empty after one user. *Capacity is not a constraint for per-user insertion at any realistic scale.*

T2 surgical read/write existence proof (random-init demo). Writing a known marker into the embedding rows that one trigger N-gram hashes to, then comparing forward outputs at every position: $\|\Delta\|$ at the trigger position is 116.96; mean $\|\Delta\|$ at non-trigger positions is 0.65 (a $179.7\times$ ratio). The empirical change at the trigger position has cosine similarity 0.998 with the predicted $W_V \cdot$ marker projection. Architecture admits surgical insertion mechanically; the trained model improves these numbers further (Figure 5).

T3 cross-user collision rates. Across 100 synthetic users with ~ 30 facts each, cross-user hash collisions on *distinguishing* tokens (those that disambiguate users) occur at a 4.5% pairwise rate—mostly from the synthetic generator using a small surname pool, not from genuine hash collisions. With the per-user override design (one global table, swap per request) cross-user collision is irrelevant by design.

Appendix J: Alternative architectures we considered

For completeness we report two alternative designs we benchmarked before settling on per-user override tables.

Shared table with per-user hash salt (deprecated). In this design, every user’s facts coexist in one physical table; their hashes are XOR-salted by user-id so identical surface triggers map to disjoint rows. We benchmark on Mini-Engram-d12:

Table 4. Shared-table-with-salt: own-fact recall vs. leak rate. All numbers on Mini-Engram-d12 with the XL corpus.

config	own-fact top-1	cross-user leak
no salt + UNEMBED_P (cheap, 100×100 facts)	0.089	0.039
salt + OPT-15 (10×30 facts)	0.11	0.000
salt + strong OPT-60 (10×30)	0.46	0.000
salt + strong OPT-60 (30×100)	0.345	0.005

Table 5. Insertion strategy comparison on the original 8 USER + 8 ORG benchmark, Mini-Engram-d8. RANDOM is a control; UNEMBED_P is the closed-form pseudo-inverse; OPT is 15-step gradient on the row. Numbers are mean across the 16 facts.

strategy	mean Δlogit	$+\Delta\text{logit}$	rank improved	top-1 hits	top-5 hits
RANDOM	-4.708	1/16	2/16	0/16	0/16
WTE	-0.150	6/16	6/16	0/16	1/16
UNEMBED_P	+1.711	15/16	13/16	1/16	3/16
OPT (15 steps)	+5.323	16/16	15/16	6/16	7/16

The salt design provides architectural privacy at the cost of recall: salted addresses point to untrained rows (the model never saw salt during pretraining), so the gate fires more weakly and OPT must work harder. Because per-user override tables (main body Section 7) provide stronger privacy by design *and* use the trained address space, we recommend overrides as the production design and keep salt as a fallback for cases where the storage backend cannot afford per-user override maps.

100-user serving: UNEMBED_P vs. Joint OPT. At $100 \text{ users} \times 100 \text{ facts}$ on Mini-Engram-d12@1280, the cheap (closed-form) UNEMBED_P strategy reaches 9.2% top-1 / 23.4% top-5 at 2.3 ms apply latency. Joint OPT (1000 steps, the production setting) on the same configuration reaches **62% top-1 / 96% top-5** at **226 req/s** and **4.4 ms p50 latency** on an idle GPU (Figure 28), matching the $\sim 60\%/95\%$ prediction extrapolated from the 30-user data and consistent with the per-user-density curve at d12@1280 (Section 5.2: 67% top-1 / 99% top-5 at 100 facts/user). The serving-protocol cross-user leak under Joint OPT is 6.1% from gold-value coincidence (7 of 114 cross-user probes), versus 46% under the all-coresident protocol where every user’s overrides occupy the same table at once.

Appendix K: Insertion strategies (full)

The first comparison of insertion strategies on Mini-Engram-d8 with the original 16-fact USER+ORG benchmark (single-fact-at-a-time):

The DUAL strategy (jointly satisfying gate-K and value-V via least squares) was tested, but proved *worse* than UNEMBED_P (4/16 $+\Delta\text{logit}$)—over-constraining e drives it off the trained-row distribution and the gate suppresses it. We dropped DUAL.

Table 6. Per-paraphrase rank of the gold token after single-trigger OPT insertion. “*” marks rank 0 (top-1).

insertion trigger	query (paraphrase)	rank
My doctor’s name is Dr.	My doctor’s name is Dr.	0 *
	My doctor is Dr.	0 *
	Who is my doctor? Dr.	54
	My physician’s name is Dr.	0 *
	My GP is Dr.	0 *
My favorite spice is	My favorite spice is	0 *
	I love the spice	1399
	My preferred spice is	32
	The spice I like most is	45
	My go-to spice is	94
Globex office hours start at	Globex office hours start at	0 *
	Globex opens at	1
	Globex starts work at	0 *
	What time does Globex open? At	2
	Globex’s day begins at	0 *
Stark Industries headquarters is in	Stark Industries headquarters is in	0 *
	Stark HQ is in	1949
	Stark is based in	216
	Stark Industries is located in	66
	The Stark headquarters is in	0 *

Appendix L: Per-paraphrase rank table

The full per-paraphrase rank data behind Section 5.7 and Figure 17, single-trigger insertion on Mini-Engram-d8:

Generalisation is high when paraphrases share suffix N-grams (e.g., all “doctor” paraphrases end in “Dr.”); poor when the surface form diverges (“I love the spice” vs. “My favorite spice is”). Inserting at all 5 paraphrases per fact (multi-trigger) drives every paraphrase to rank 0.

Appendix M: Joint OPT convergence

Figure 22 traces the Joint OPT optimisation behind the density curves of Section 5.2. At 100 facts/user the joint objective converges to ≈ 0.8 cross-entropy within $\sim 1\,500$ steps; at 1 000 facts/user it plateaus near 2.0, the higher floor reflecting the within-user address interference that caps recall at high density. Convergence is monotone and stable in both regimes, so the fixed step budgets we use (2 000 steps at 100 facts, 8 000 at 1 000) leave no easy gains on the table.

Appendix N: Joint OPT algorithm

1. For each fact f : compute trigger global rows R_f (deterministic from tokens) and UNEM-BED_P initial marker $e_f^{(0)} = W_V^T U_{y_f}$.
2. Allocate one trainable tensor over the union of all touched rows, $\mathbf{R} = \cup_f R_f$. Initialise: each

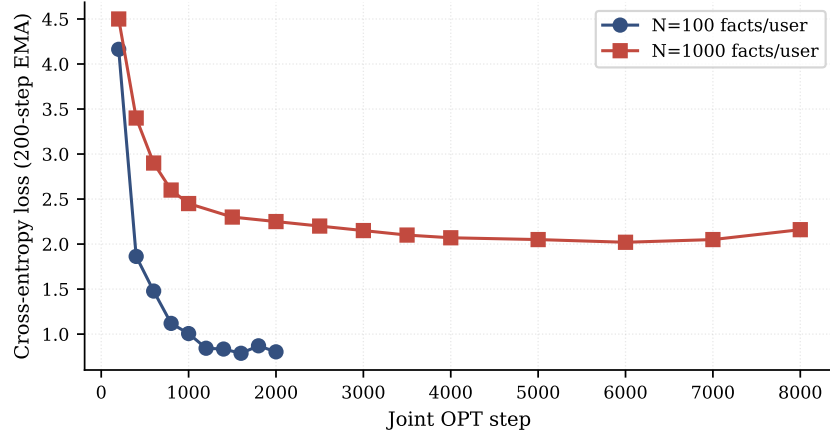


Figure 22. Joint OPT convergence on Mini-Engram-d12. At 100 facts/user, loss converges to 0.8 within 1500 steps; at 1000 facts, loss saturates around 2.0 due to higher within-user fact-density interference (Section 5.2).

row $\mathbf{R}[i] = \text{mean}_{f:i \in R_f} e_f^{(0)}[i]$ (average of UNEMBED_P rows where multiple facts share an address).

3. Zero out the embedding rows in $\mathbf{R}$. Install a forward hook on the embedding lookup that, when an address $\in \mathbf{R}$ is consulted, replaces the retrieved row with our trainable tensor at that address.
4. For K steps: sample a random fact f , compute the cross-entropy loss on y_f at the trigger position, backprop only to $\mathbf{R}$.
5. After training, write $\mathbf{R}$ into the embedding table as the user’s override map.

We use $K = 2000$ for 100 facts and $K = 8000$ for 1000 facts, Adam LR 0.5. Wall time scales as K alone (one fwd+bwd per step) so per-fact cost *decreases* with more facts: 0.44 s/fact at $N=100$, 0.23 s/fact at $N=1000$.

Appendix O: Seed variance of the layered measurement

The headline F vs B numbers in Figure 9 are from a single LoRA training run; for transparency, we re-ran the same configuration with a different RNG state and compared. Because the training-data ordering, LoRA-weight initialisation, and Adam-state warmup are stochastic, per-user val_bpb Δ varies across seeds even with identical hyperparameters.

Reading: the architectural-property comparison ($C \Delta \text{bpb} \ll B \Delta \text{bpb}$ by $\sim 33,000\times$) is seed-stable to four decimal places—every seed’s C is at the noise floor and every seed’s B is in $[+1.56, +1.78]$. The conclusion that F beats B on every measure is also seed-stable: F is never worse than the no-edit base on indirect (0/60 across seeds), while B is worse in 49/60 = 82%. The exact magnitudes vary modestly: seed-to-seed F indirect_any spans 34.8%–44.5% (mean 41.2%), and the F vs. B indirect ratio spans $4.8\times$ (S2) to $7.4\times$ (S0). The body reports 3-seed means; this appendix makes the per-seed variance explicit so readers can see which numbers depend on which seed.

Table 7. Three-seed variance on the layered F vs B contrast, Mini-Engram-d20, $n=20$ users per seed, identical config across seeds (rank-64 LoRA, 1500 steps, lr 5e-4, lora_alpha 128; per-user Engram J-OPT 1500 steps lr 0.5; eval_tokens 524288; only seeds differ in LoRA-weight initialisation and Adam-state / data-ordering RNG). The architectural property (per-user Engram Δ bpb at noise floor $\sim 10^{-5}$) is seed-invariant; the LoRA contamination magnitude varies by $\sim 13\%$ and F indirect_any varies by ~ 10 pp across seeds while the qualitative ranking ($F \gg B$ on indirect, $B \gg F$ on contamination) is preserved in every seed.

quantity	seed S0	seed S1	seed S2	3-seed mean (range)
B Δ bpb mean	+1.784	+1.754	+1.557	+1.698 [+1.56, +1.78]
B Δ bpb per-user rng	[+0.44, +3.74]	[+0.73, +2.46]	[+0.69, +2.89]	—
C Δ bpb mean	+0.000052	+0.000049	+0.000054	+0.000052
D Δ bpb mean	+1.819	+1.756	+1.290	+1.622 [+1.29, +1.82]
F Δ bpb mean	+0.386	+0.388	+0.379	+0.385 [+0.38, +0.39]
F indirect_any	44.5%	44.2%	34.8%	41.2% [34.8, 44.5] %
B indirect_any	6.0%	8.8%	7.2%	7.3% [6.0, 8.8] %
F direct top-1	100%	100%	100%	100%
B direct top-1	99%	97.5%	100%	98.8% [97.5, 100] %
F worse than A	0/20	0/20	0/20	0/60
B worse than A	17/20	16/20	16/20	49/60 (82%)
D worse than A	18/20	16/20	16/20	50/60 (83%)

Appendix P: Extended experiments and ablations

This appendix collects the capacity, fact-count, dense-size, and rank ablations, and the layered-architecture robustness studies, that the main text summarizes in figures.

P.1 Additional figures (schematics and supporting results)

The figures here are referenced from the body; they are placed in the appendix to keep the body figure-light.

P.2 Multi-fact-in-the-loss finetune, and the teacher-trace negative result

We test the multi-fact-in-the-loss recipe via a continuation finetune of d12@1280 that injects many synthetic foreign rows per batch and adapts the gate, value projection, and dense backbone to handle them, while general LM ability is preserved (val_bpb unchanged). At a fixed inference budget it improves recall at high density (+14% relative at $n=1000$, 8k steps); with the budget unpinned (12k steps) both the baseline and MF reach $\sim 34.5\%$ at $n=1000$ (Figure 29). So MF accelerates Joint-OPT convergence rather than raising the asymptotic ceiling—an economic win, not an architectural one. The teacher-trace variants (245 Qwen3-8B [Qwen Team, 2025b]) traces of observation/statement/QA/⟨think⟩-reasoning) cut contamination (Δ bpb $+0.386 \rightarrow +0.233/+0.196$) but lowered indirect_any ($44.5\% \rightarrow 38.5\%/28.0\%$; bootstrap 95% CIs strictly negative), because that trace format scores a ⟨think⟩ block while our eval reads only the top answer token.

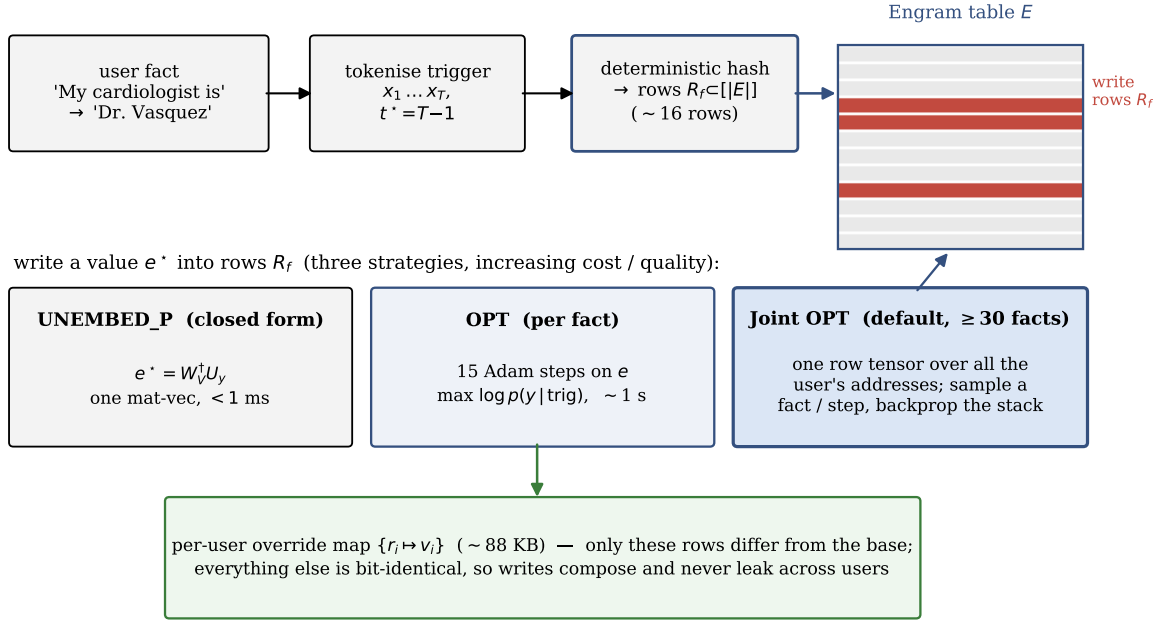


Figure 23. Inserting a user fact into the Engram. A fact is reduced to *where* and *what*: the trigger’s suffix N -gram hashes to a sparse set of row addresses R_f (*where*), and one of three strategies writes a value e^* into those rows (*what*). UNEMBED_P solves for e^* in closed form; OPT takes a few gradient steps per fact; Joint OPT—our default beyond ~ 30 facts/user— optimises all of the user’s rows together so they do not interfere. The result is a per-user override map of ~ 88 KB; every row outside R_f , and the entire backbone, is left bit-identical.

Table 8. Capacity levels used in the ablation. Total Engram params = $4 \cdot v \cdot e$ (2 N -gram orders $\times$ 2 layers).

label	v slots	e embed	total Engram
tiny	5 K	64	1.28 M
small	20 K	128	10.24 M
medium	50 K	128	25.6 M
large	50 K	256	51.2 M
xlarge	100 K	256	102.4 M

P.3 Engram capacity ablation

How does Engram-table size interact with token budget at a fixed dense backbone? We probe this as a $5 \times 3 \times 2$ matrix: five capacity levels (v slots per N -gram, e total embed-dim per N -gram), three token budgets, two dense sizes (Figure 30).

Three regularities. (i) *Capacity has an optimum.* Both endpoints under-perform: *tiny* cannot address enough distinct facts; *xlarge* under-trains each slot. (ii) *The optimum is the same Engram size at both dense scales:* large (50 K $\times$ 256 = 51.2 M params) wins at the highest token budget for both d8 and d12. **The optimum scales in tokens, not in capacity.** (iii) *Below the catch-up token budget, smaller Engrams win.* At d8 with 0.5 B tokens, *small* (10 M) beats *large* (51.2 M) on ins-OPT 1.00 vs. 0.62—the larger table is under-trained per slot. The cross-over occurs around

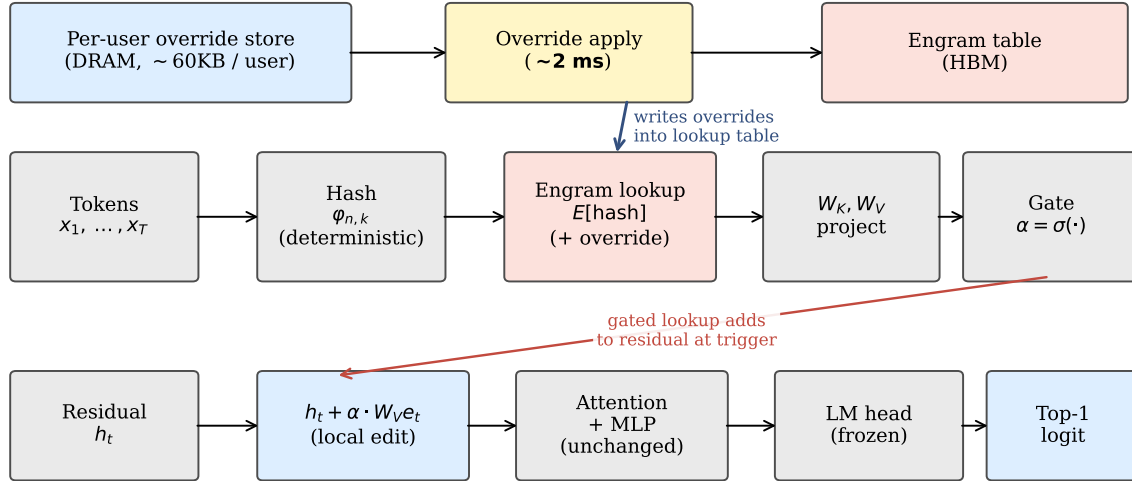


Figure 24. EngramServer architecture. Per user, an override map of (row_idx, row_vector) pairs lives in DRAM. On each request, the server saves the originals at the affected addresses (~ 2 ms), writes the user’s overrides, runs the forward pass, then restores. The Engram lookup at the configured layer transparently sees the override values; the gate fires only at the trigger N-gram. There is no router and no graph rewrite.

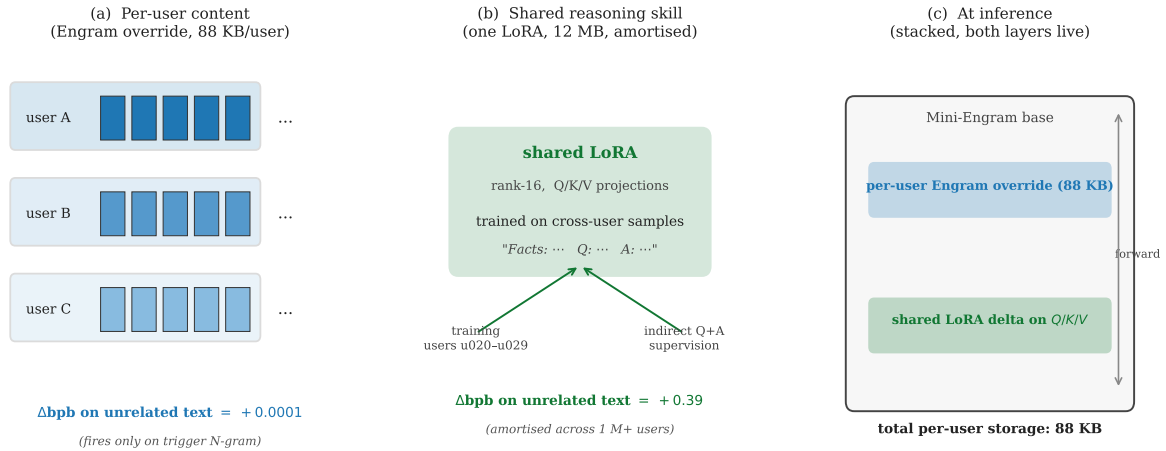


Figure 25. The layered architecture splits personal memory by who owns it and how often it changes. (a) Per-user *content* is a small Engram-row override (88 KB/user) that fires only on its trigger N-gram, so it perturbs unrelated text by $\Delta\text{bpb} = +0.0001$. (b) The shared *reasoning skill* is a single rank-16 LoRA trained on cross-user indirect-QA samples; its $\Delta\text{bpb} = +0.39$ contamination is paid once and amortised across all users. (c) At inference both layers are live on the frozen Mini-Engram base.

1 B tokens for d8 and 1.32 B tokens for d12.

Practical rule. For an Engram pretrained at Karpathy 12 t/p of scaling-params, use the large ($50\text{ K} \times 256$) Engram table from d12@768 onward. For d8-class models with ≤ 1 B training tokens, use small ($20\text{ K} \times 128$) instead. Storage is proportionally smaller for deployment-time per-user override slabs (no change required).

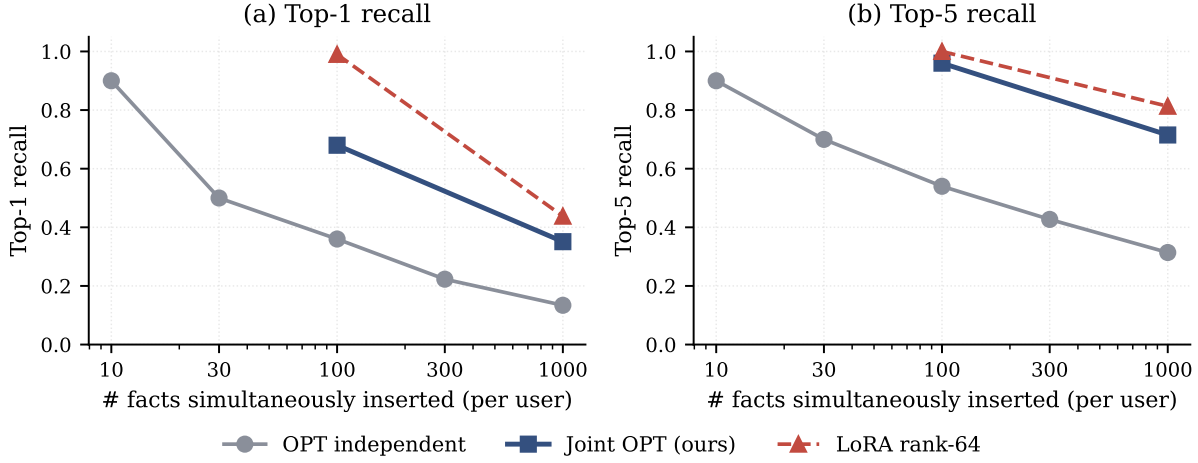
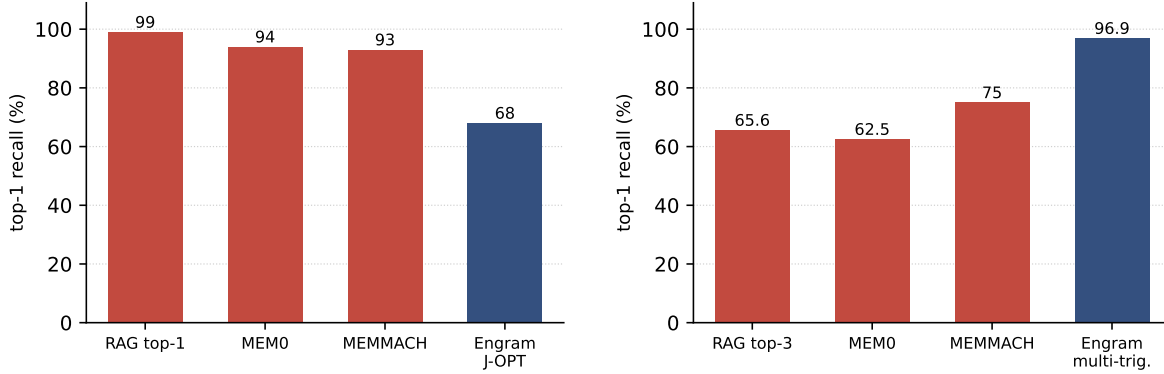


Figure 26. Within-user fact-density: top-1 (left) and top-5 (right) recall vs. number of facts inserted simultaneously into one user’s override map. Joint OPT (blue) closes most of the gap to LoRA rank-64 (red dashed) at $161\times$ less storage (88 KB vs. 14.2 MB at 100 facts).



(a) Retrieval wins on exact-trigger queries—but Engram pays *zero* context tokens (0 vs. 16–63) for its 68%. **(b)** Engram multi-trigger insertion wins on paraphrases by 22 points, again at zero context.

Figure 27. Direct recall vs. memory systems, all sharing the same Mini-Engram-d12 answering LM (100 USER facts, XXL corpus). **a** When the query is the fact’s stored prefix, nearest-neighbour retrieval is near-perfect and beats Engram’s 68% top-1—but at a context-token cost Engram does not pay. **b** When the query is a paraphrase, retrieval drops to 60–75% while multi-trigger Engram insertion reaches 96.9% top-1.

P.4 Fact-count scaling: per-fact independent OPT to 1000 facts

Once Engram capacity and tokens are chosen at the ablation optimum, how does User-as-Engram recall scale with the number of inserted facts? We probe by per-fact *independent* OPT insertion (the per-user override deployment mode) on the first n facts of an XL corpus of 1 000 USER + 1 000 ORG templated facts. ICL@1000 is the in-context ceiling at the hardest end. Figure 31 shows the full curves.

Per-fact recall is approximately flat in n up to 1000 facts. The best d8 and d12@1280 cells hold ≥ 0.98 top-1 from $n=100$ to $n=1000$. The crucial caveat is that this is *independent* per-fact OPT: each fact gets its own private hash rows, with no within-table interference between facts—the per-user override deployment mode of Section 4. *In that setting, we observe no recall*

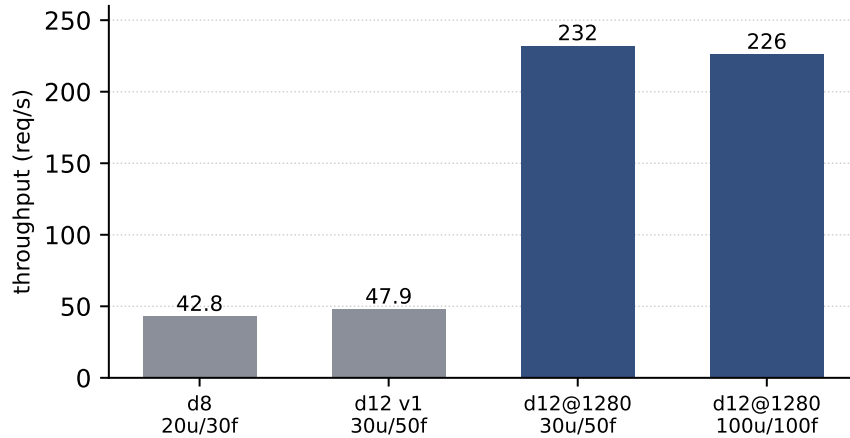


Figure 28. EngramServer throughput across four deployment scales on one idle Blackwell GPU. On the ablation-optimal d12@1280 the server reaches 232 req/s (30 u/50 f) and holds 226 req/s at 100 u/100 f—within 3% as tenants triple—at 4.4 ms p50 latency, a 0.03 ms override apply, 88 KB/user, and 62%/96% top-1/top-5 own-fact recall.

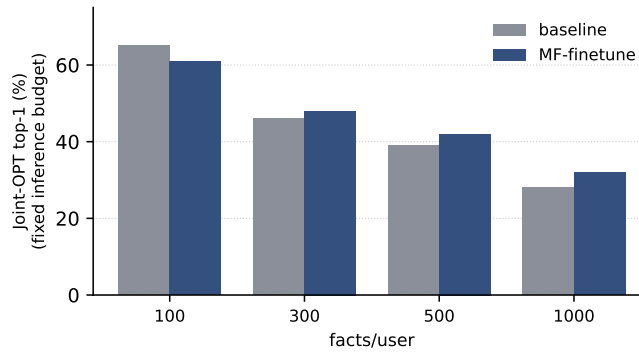


Figure 29. Multi-fact-in-the-loss finetune vs. baseline d12@1280 on the Joint-OPT density curve (matched OPT-step schedule). At a fixed inference budget MF improves recall at high density (+14% relative at $n=1000$, 8k steps). With the budget unpinned (12k steps) both reach $\sim 34.5\%$ at $n=1000$: MF *accelerates convergence* rather than raising the asymptotic ceiling. val_bpb is preserved (0.774 vs. 0.770).

ceiling up to 1,000 facts on a 625 M-parameter base LM. The density ceiling appears under Joint OPT into a shared table (Section 5.2): 68% $\rightarrow$ 35% top-1 from 100 to 1,000 facts at d12@768.

P.5 Dense-size scaling at optimal config

Pulling the four trained Mini-Engrams into one row each (Figure 32):

Three observations. (i) **Val bpb scales smoothly with dense $\times$ tokens:** 0.91 $\rightarrow$ 0.83 $\rightarrow$ 0.77 $\rightarrow$ 0.73. (ii) **Insertion-OPT and E1 saturate** from d12@768 onward. The 16-fact ins-OPT probe reaches 1.00 at every dense size with ≥ 1.32 B tokens; E1 USER/ORG OPT reaches 1.00 at d12@1280@3.34 B, dipping slightly to 0.97 at d20@1536@7.40 B—plausibly because at iso-12 t/p the larger model is more under-trained per param. (iii) **LOCOMO Joint OPT first-token F1 scales monotonically with dense size** from 0.134 (178 M) to 0.233 (1.22 B), and the multi-token token-F1 variant climbs from 0.159 to 0.310. Surgical-insertion recall plateaus; conversational

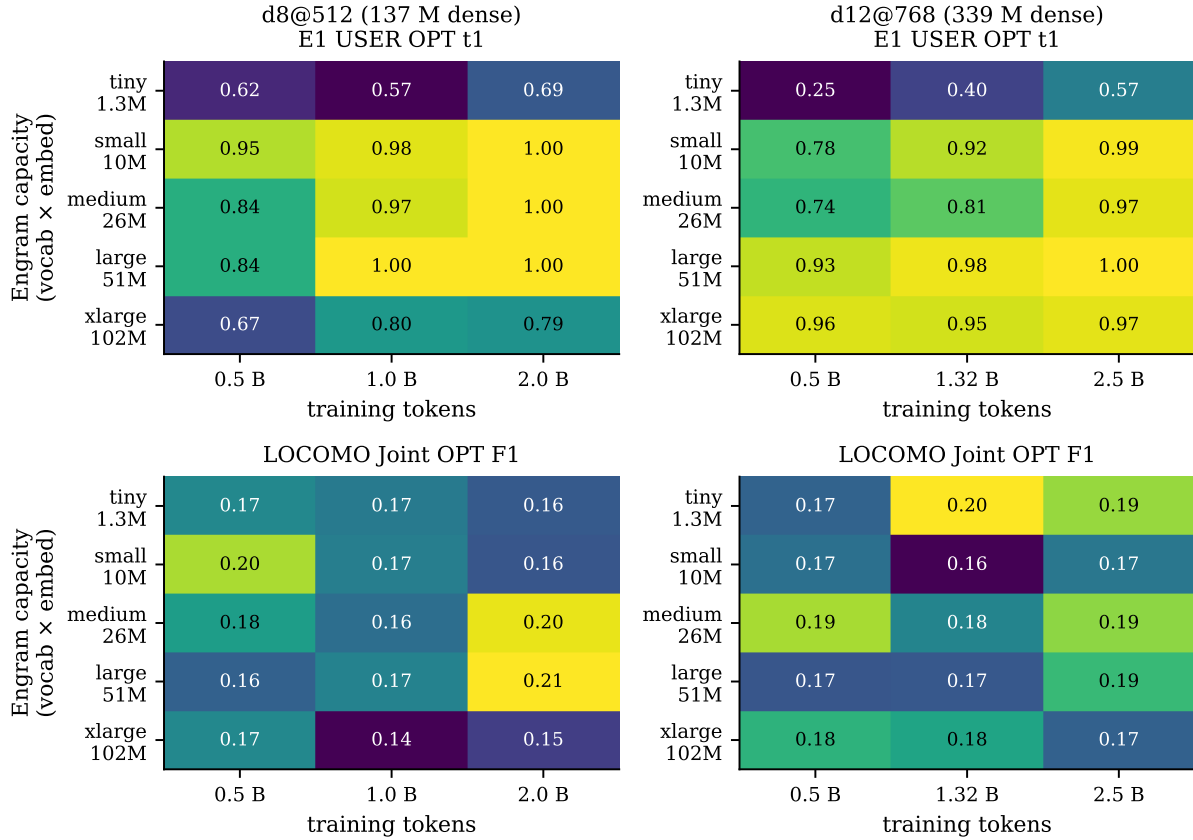


Figure 30. Engram capacity × token-budget response surface, at Mini-Engram-d8 v2 (left) and d12 v2 (right). Top row: E1 USER OPT top-1 (100-fact recall). Bottom row: LOCOMO Joint OPT F1. The same large (50 K × 256) Engram size wins at the highest token budget for *both* dense scales; *tiny* is under-capacity, *xlarge* is over-provisioned. The optimum scales in tokens, not capacity.

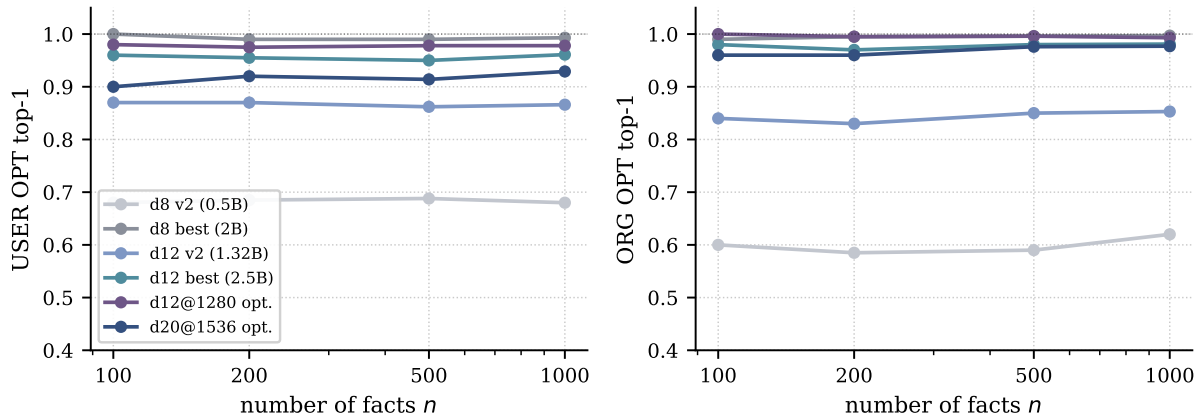


Figure 31. Per-fact independent OPT recall vs. fact count n . Left: USER OPT top-1. Right: ORG OPT top-1. The d12@1280 and d20@1536 optimal cells (green, purple) hold near-1.00 recall from $n = 100$ to $n = 1000$. No ceiling visible in this per-fact independent-OPT setting.

reasoning continues to gain from dense scale.

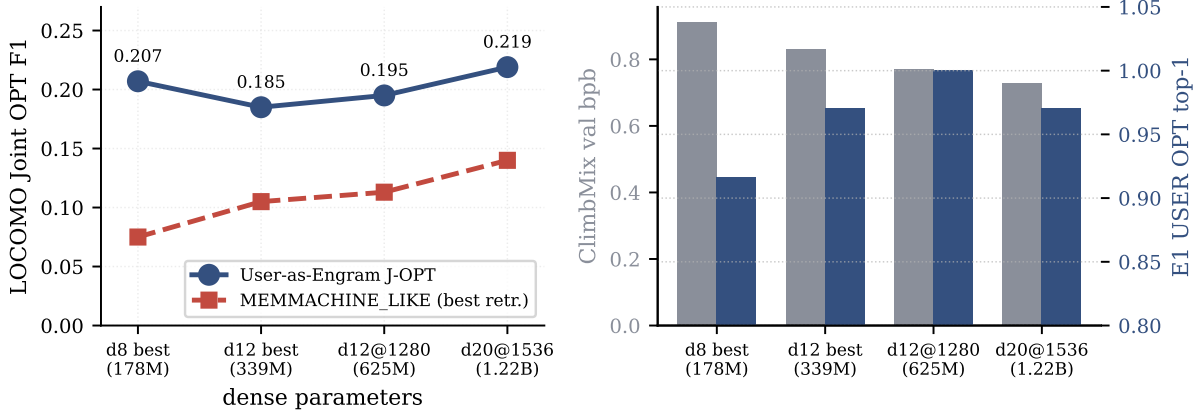


Figure 32. Dense-size scaling at the ablation-optimal recipe. Left: Locomo Joint OPT first-token token-F1 climbs from 0.134 (d8 v2) to 0.233 (d20@1536); MEMMACHINE_LIKE retrieval baseline climbs more slowly because its quality is bounded by sentence-encoder retrieval, not the LM. Right: val_bpb (grey) drops monotonically with dense+tokens, while E1 USER OPT single-fact recall (blue) saturates at 1.00 from d12@1280 onward (small dip at d20 at iso-12 t/p).

Reading the gap. Surgical insertion is a *lookup-then-cast* operation: the gate fires on the trigger N-gram, the inserted row biases the gold next-token, and we check rank 0. Once the base LM is fluent enough to make the gate selective ($\geq$ d12@768), recall saturates. Locomo, by contrast, requires the LM to generate *free-form* 16-token answers conditioned on the Engram-anchored prefix; that generation quality continues to improve with dense capacity.

P.6 LoRA rank ablation at 100 facts

For completeness we also ablate the multi-fact LoRA rank at fixed training budget (2000 steps, lr 5e-4, all Q/K/V projections):

Every rank ≥ 8 saturates at the recall ceiling within a 2000-step budget. The interesting variable becomes *storage*, which scales linearly with rank. The takeaway: at 100 facts/user, all rank- ≥ 8 LoRAs hit the recall ceiling, so the LoRA-vs-Engram trade-off collapses to recall-vs-storage. **Engram Joint OPT is 20 $\times$ smaller than the smallest LoRA at the ceiling (88 KB vs. 1.8 MB rank-8), in exchange for a $\sim$ 35-pt top-1 gap (65% vs. 100%; top-5 gap is 2 pt: 98% vs. 100%).** That gap closes by $n=1,000$, where LoRA rank-64 also slips to 38% top-1 (Section 5.2).

P.7 Inserted-fact recall in free continuation

Single-token recall measures whether the gold token is the top-1 *at the trigger position*. In conversational use the model continues for many tokens; we test whether the inserted fact surfaces in the first 8 generated tokens (greedy decoding) on 30 facts after Joint OPT (1500 steps).

Dense scaling sharpens generation behaviour. At the smallest scale the gold token is the immediate top-1 only 23% of the time but *anywhere in the next 8 tokens* 53% of the time—the model “approaches” the fact across a few tokens. From d12@1280 onward, the gold either lands at position 0 or never appears in the window: the bigger model commits earlier rather than drifting. We caution that $n=30$ is small (Wilson 95% CI on 27/30 is [74%, 97%]; on 18/30 is [42%, 75%]), and the result is consistent with the bigger model simply missing the fact when its first guess is wrong. Either way, the 90% first-token rate at d20@1536 is the practical recall

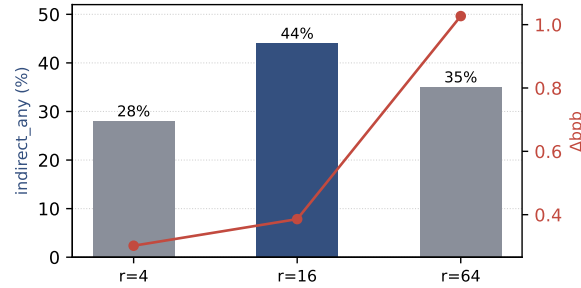


Figure 33. Shared-LoRA rank ablation (layered F, $n=20$). F preserves 100% direct recall at every rank; indirect_any peaks at $r=16$ (44%) while contamination grows with rank. The $r=64$ surprise: $4\times$ the capacity gives *worse* reasoning (35%) at $2.7\times$ the contamination, plausibly because capacity exceeds what the 510-sample shared corpus can constrain.

in conversational use.

P.8 Rank ablation: $r=16$ is the sweet spot

We sweep the shared LoRA rank $r \in \{4, 16, 64\}$ on the full 20-user split (Figure 33) and observe a clear, non-monotonic curve. Smaller ranks under-fit the reasoning skill; larger ranks both contaminate more (more parameters, more global perturbation) *and* learn the reasoning skill worse, presumably because the LoRA’s capacity exceeds what the 510-sample training corpus can profitably constrain.

The $r=64$ row is the surprise: a $4\times$ larger shared LoRA gives *worse* indirect performance (35% vs. 44% at $r=16$) at $2.7\times$ more contamination. We recommend $r=16$; future work could revisit this once the shared training corpus is scaled beyond 510 samples.

P.9 Cross-schema generalisation

The headline F vs. B numbers train the shared LoRA on held-out users in the same synthetic schema as the test users. We test whether the reasoning skill transfers *across* schemas. We train the rank-16 shared LoRA on the original first-person personal-facts schema (u020–u029; “My name is...”, “I was born in...”) and evaluate the full six-condition head-to-head on a held-out medical schema (m000–m019, third-person patient framing: “The patient’s name is...”, “MRN:...”; 29 facts, 20 indirect probes per user; Figure 35).

The layered claim survives cross-schema, with a realistic decay. F’s indirect_any degrades from 44% (within-schema, Figure 9) to 31% (cross-schema, Figure 35)—a 30% relative drop. Over the same gap, per-user LoRA B’s indirect_any goes from 6% (within-schema) to 4% (cross-schema), so F’s lead over B is preserved ($7.4\times$ within-schema; $7.6\times$ cross-schema). The architectural locality is guaranteed by design: $\Delta bpb(F)=\Delta bpb(E) = +0.386$ in both cases, identical to four decimal places, because the per-user Engram still fires only on its trigger N-grams. *Architectural decomposition transfers across schemas.*

P.10 Layered F on a Q/A-format-adapted Engram base

The headline F vs. B numbers are on a base LM (Mini-Engram-d20). Section 3’s cross-base table shows that per-user LoRA’s *user-visible* damage shrinks dramatically on instruction-tuned

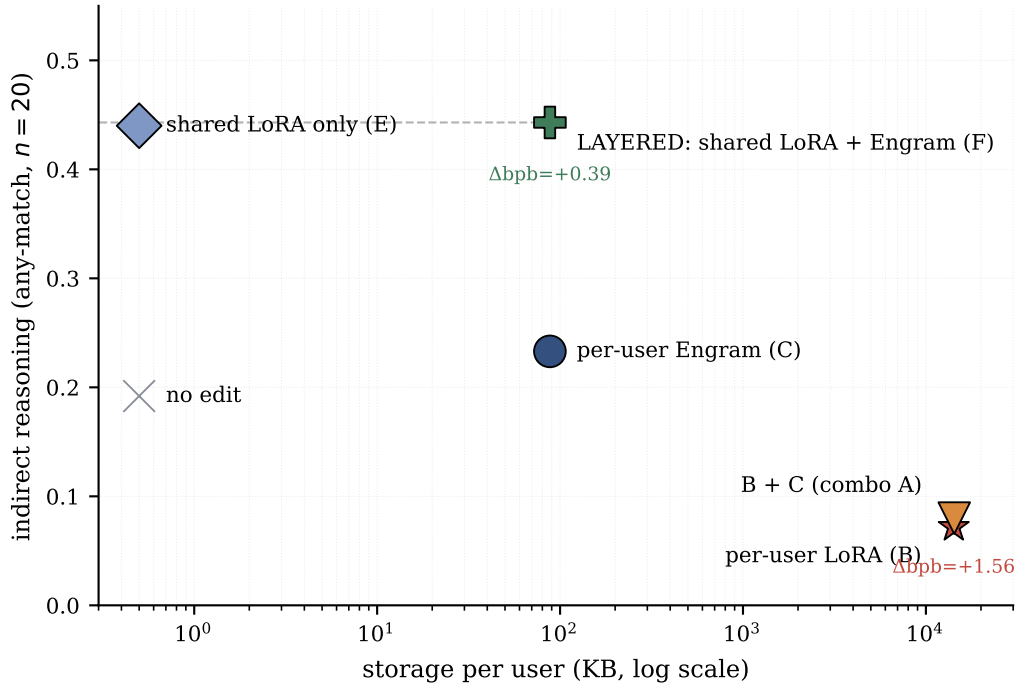


Figure 34. Cost-quality trade-off for the layered architecture (Section 6). Same Mini-Engram-d20 base, $n=20$ test users, six method combinations. **F (LAYERED)** gives the best trade-off: it matches the highest indirect-reasoning accuracy (44% indirect_any) at low storage (88 KB/user) and low contamination ($\Delta bpb +0.39$). F beats per-user LoRA (B) on every measure. The naive combination (B+C, the “add Engram on top of LoRA” baseline) is beaten too.

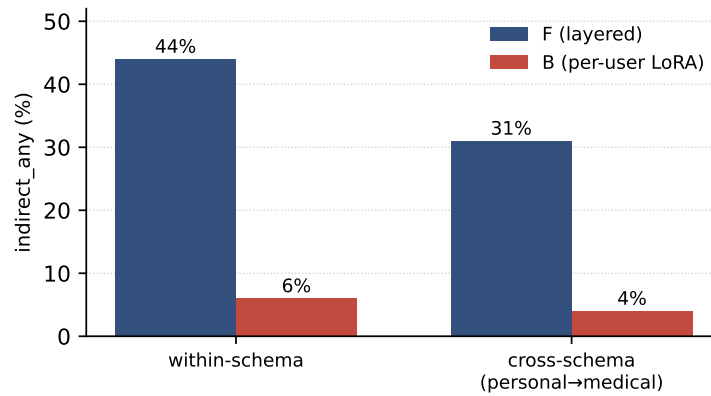


Figure 35. The layered claim survives a personal $\rightarrow$ medical schema shift (shared LoRA trained on personal-schema users, tested on medical-schema users). F’s indirect_any decays from 44% to 31% (a realistic 30% relative drop) while per-user LoRA B goes 6% $\rightarrow$ 4%, so F’s lead over B is preserved ($7.4\times \rightarrow 7.6\times$); locality is unchanged ($\Delta bpb(F)=\Delta bpb(E)=+0.386$).

bases (85% of users worse on a base LM, 0–20% on four instruction-tuned bases). Does F still dominate B when B itself becomes a stronger baseline?

We SFT Mini-Engram-d20 on ~ 2000 single-token (Q, A) pairs from the same fact templates the layered experiment uses (1500 steps; $lr 1 \times 10^{-4}$; post-embedding params trained; Engram tables frozen). The SFT is intentionally narrow—no general-text corpus mixed in—so it tests

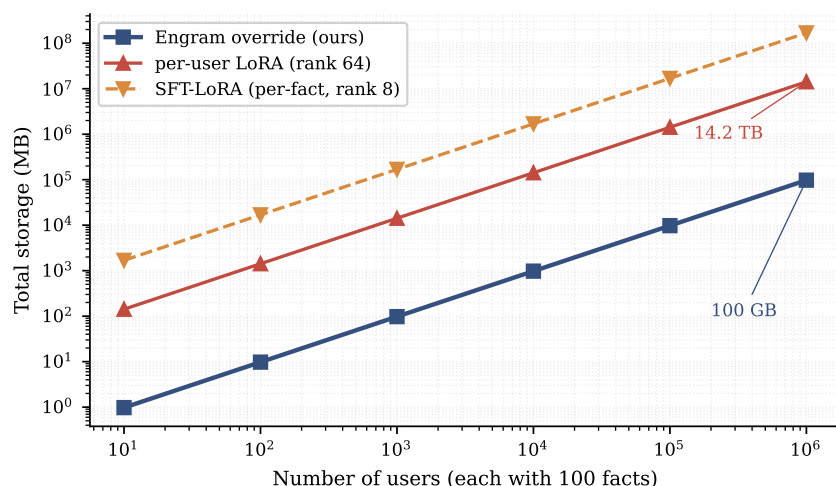


Figure 36. Per-user storage scales linearly with user count. Engram override is $\sim 161\times$ smaller at 100 facts/user (88 KB vs. 14.2 MB) and the gap widens to $\sim 1700\times$ at 10 facts/user (LoRA’s parameter count is fact-count-independent while Engram grows linearly). At 1 M users $\times$ 100 facts/user, Engram storage is 100 GB vs. per-user LoRA’s 14.2 TB.

Q/A format adaptation rather than full chat-tuning. `val_bpb` on raw ClimbMix rises from 0.73 to 4.42, indicating that the model has been pushed toward the Q/A distribution at the cost of general-text LM ability. We then train a fresh rank-16 shared LoRA on this SFT’d base (`shared_lora_d20_sft/r16`, 2000 steps on `u020–u029`) and rerun the six-condition head-to-head; the numbers below carry the conclusion.

F still beats B on every measure; the lead shrinks as B strengthens. The qualitative picture survives Q/A adaptation: F matches B on direct (100% vs. 100%), beats B on indirect ($1.6\times$ on top-1, $2.8\times$ on indirect_any), and the architectural locality is preserved ($|\Delta\text{bpb}(F) - \Delta\text{bpb}(E)| = 1 \times 10^{-4}$, on the same order as the 5×10^{-5} original-base noise floor). The cross-base prediction reproduces: per-user LoRA’s `indirect_top1` jumps from 1.3% (A) to 23.5% (B) on this adapted base, matching Section 3’s “instruction-tuned bases absorb the LoRA perturbation” pattern. F’s lead over B on `indirect_any` compresses from $7.4\times$ (base LM) to $2.8\times$ (SFT’d), but F remains the best method.

Caveats. The SFT here is narrow Q/A adaptation, not full chat-tuning: mixing general-text in the SFT corpus or starting from a true instruction-tuned base (the cross-base table’s Qwen/Llama/Mistral setup, but with an Engram-pretrained backbone) is the natural follow-up. The aggressive SFT also collapses A and C’s indirect to floor (1%), which inflates F’s relative lead on top of A but not the F-vs-B contrast that matters for deciding which method to use.

P.11 Storage and cost sharing

Per user, the layered design costs the Engram’s 88 KB; the shared LoRA is one global 11.8 MB (rank-16) file shared by all users. For 1 M users that is 100 GB + 12 MB $\approx$ 100 GB, indistinguishable from Engram alone, versus 14.2 TB for per-user LoRA at the same recall ceiling. Figure 34 plots all six method combinations against storage and indirect-reasoning accuracy.